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A Bayesian workflow for securitizing casualty insurance risk

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Abstract

Casualty insurance-linked securities (ILS) are appealing to investors because the underlying insurance claims, which are directly related to resulting security performance, are uncorrelated with most other asset classes. Conversely, casualty ILS are appealing to insurers as an efficient capital management tool. However, securitizing casualty insurance risk is non-trivial, as it requires forecasting loss ratios for pools of insurance policies that have not yet been written, in addition to estimating how the underlying losses will develop over time within future accident years. In this paper, we lay out a Bayesian workflow that tackles these complexities by using: (1) theoretically informed time-series and state-space models to capture how loss ratios develop and change over time; (2) historic industry data to inform prior distributions of models fit to individual programs; (3) stacking to combine loss ratio predictions from candidate models, and (4) both prior predictive simulations and simulation-based calibration to aid model specification. Using historic Schedule P filings, we then show how our proposed Bayesian workflow can be used to assess and compare models across a variety of key model performance metrics evaluated on future accident year losses.

Keywords: bayesian workflow, insurance-linked securities, loss development, loss forecasting, casualty insurance

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1 Introduction

Insurance-linked securities (ILS) are financial instruments that enable insurers to transfer risk to the capital markets. Traditionally used in the context of catastrophe insurance risk (e.g. AAA Extreme Events and Property Lines Committee, 2022; Lane, 2024), ILS have arisen as effective tools for capital management, providing insurers with sources of capacity from outside the traditional reinsurance industry and investors with access to a novel asset class that is relatively uncorrelated with other assets in the marketplace (Jaeger et al., 2010).

The insurance market is often described as cyclical, where profits grow and fall periodically as underwriters observe and respond to both micro- and macro-economic factors that impact program performance (e.g. Berger, 1988). Although the existence of true, predictable periodicity in insurance underwriting is debated, periods where premiums become higher or lower relative to losses are readily observed (e.g. Boyer et al., 2012). In recent years, "hard" market conditions have lead to a significant increase in attention given to the problem of securitizing casualty insurance risks. The reason is straightforward – individual casualty insurers, and subsequently the casualty insurance market as a whole, can only hold a limited amount of risk. As capacity for risk reaches the limit, the ability to write new risk and pursue new opportunities diminishes, resulting in stagnated growth. In such conditions, casualty insurers need to source capital efficiently so that they can take advantage of opportunities while not diluting existing shareholders through other capital management strategies. ILS provides exactly this flexibility, allowing insurers to release capital otherwise locked in long-tailed reserves and then redeploy it in other areas (see Canabarro et al., 2000). This ability to access external capital allows insurers to compete more effectively, control more business than their balance sheets would typically support, and signal the quality of their underwriting through favorable terms from ILS investors.

Despite the promise of casualty ILS, there are significant technical challenges involved in securitizing the underlying casualty insurnace risk. First and foremost, casualty insurance is notoriously "long-tailed", meaning that it can take years or even decades for claims to be settled depending on the specific line of business. For example, for worker's compensation polices written in 2024, we may not known the losses incurred by associated claims until 2034, 2044, or even later depending on the nature of the risk (CAS Tail Factor Working Party, 2013). This long-tailed behavior means that the true losses associated with a casualty insurance portfolio are often not known for years down the road, presenting a significant challenge for pricing casualty ILS. This uncertainty is further compounded in cases where the insurance program in question has sparingly little historic data available to use for informed decision-making, as is often the case with new products seeking reinsurance or ILS opportunities.

Due to its flexibility and the ease at which it can accommodate external information, the Bayesian framework is particularly well-suited to address the complexity and uncertainty associated with modeling casualty insurance risk. Bayesian models are used across a diverse range of industries to address exactly these issues, including in pharmaceuticals (Lesaffre

et al., 2020), energy (Adedipe et al., 2020), marketing (Wang et al., 2017), economics and finance (Martin et al., 2023), and insurance loss reserving more generally (Meyers, 2015). Perhaps most conceptually relevant to our work, Bayesian models have also been proposed for use in valuation of catastrophe bonds (Domfeh, 2023), a form of catastrophe ILS. However, catastrophe bonds have a decades-long history of use (see Lane, 2024), and the underlying risk models are generally well-agreed upon (see Mitchell-Wallace et al., 2017). By contrast, casualty ILS has no standardized modeling framework, in part because it is a relatively novel financial instrument.

Here, we present a Bayesian workflow for modeling the long-tailed, uncertain nature of casualty insurance risk. We use both simulations and historic real-world data to show how a combination of theory-informed models, carefully-specified priors, and data-driven model averaging and selection can be used to accurately forecast and quantify uncertainty in how losses develop over time across multiple lines of business, thus making casualty ILS feasible. Our aim in presenting this workflow is to catalyze further research on casualty ILS, which we believe is essential for enhancing its practical use and promoting wider adoption by both insurers and capital providers alike.

Note that we assume basic familiarity with Bayesian analysis throughout the following sections. For readers not yet familiar with Bayesian analysis, we recommend McElreath (2018) and Gelman et al. (2013) for in-depth coverage and van de Schoot et al. (2021) for a concise primer.

2 What is a Bayesian modeling workflow?

A "Bayesian workflow" formalizes the iterative, and often non-linear, process of development, fitting, validation, comparison, and selection or averaging of Bayesian models (Gelman et al., 2020; Gabry et al., 2019). Particular workflows for Bayesian modeling can now be found for fields such as cognitive science (Schad et al., 2020) and epidemiology (Grinsztajn et al., 2021; Bouman et al., 2024), but there has been no discussion of what a Bayesian workflow should look like for actuarial science, let alone for casualty ILS modeling in particular (although Bayesian methods are gaining popularity, see De Alba, 2002; Meyers, 2015). This is particularly important for actuarial science due to the multi-stage analyses that are frequently conducted, from loss development to forecasting of future ultimate loss ratios. Enumerating a Bayesian workflow is necessary to enhance robustness and reproducibility of actuarial analyses that use Bayesian methods, including analyses needed for casualty ILS.

Below, we provide an introduction to the main components of a Bayesian workflow that are relevant to our specific workflow for modeling casualty insurance risks presented in a later section. We use a simplified Bayesian chain-ladder loss development model as a running example here, although we provide fuller illustration in Section 3.

2.1 Model development and fitting

Model development typically starts from the selection of a base model, which might be an existing model in the literature, or a newly developed model. In either case, it is important to clearly reify the the structure of relationships between observed variables and model parameters mathematically, in code, and/or graphically (Kruschke, 2021). Commonly, Bayesian models are at least partly generative (Gelman et al., 2020), specifying a joint probability distribution over the observed data and parameters, and allowing for forward simulation of new data from the data generating process.

It is helpful to consider a concrete example when describing model development. For loss development models in insurance, claims across most lines of business often take many months, or even years, to settle. For example, if a policy holder is in a car accident in November of accident year 2024, it may take a few months for their car to be repaired in an auto shop. Further, if they experienced bodily injury as a result of the accident, medical claims could persist for years into the future. When viewed in aggregate across policies (i.e. at the "program" level), this "loss development" means that the losses incurred by a pool of policies in a given accident year are not fully known until years into the future. Depending on the underlying risk, it could even take decades for losses within a given accident year to reach their "ultimate" state.

Loss development modeling is the practice of using historic patterns of loss development to predict ultimate losses for all accident years within a pool of policies. Data are typically organized into a triangular matrix of experience periods by development periods. For our purposes, the experience period is the year an accident occurs, and the development period is the number of years (or lags) since the accident year. Values in the triangle can be cumulative or incremental losses that indicate the observed loss for each accident year as of each development lag (we focus on cumulative losses here). Generally, we expect that losses within each accidenty year will asymptote to some ultimate loss as development lag increases. Some lines of business are relatively quick to develop (e.g., private passenger auto), whereas others can often take many years (e.g., worker's compensation).

One may start modeling such loss development using a Bayesian variation of the chain-ladder method (Mack, 1993; England and Verrall, 2002) as the base. The joint probability distribution of the losses, y, and parameters then might be represented as $p(y, \alpha, \sigma)$, where α is an $M-1$ vector of link ratios, one for each development lag save the first development period, and σ is the residual standard deviation. The joint density factors into the likelihood distribution, $p(y \mid \alpha, \sigma)$, and (independent) prior distributions $p(\alpha, \sigma) = p(\alpha)p(\sigma)$. The likelihood distribution for the ith accident year and jth development period would be a positive-bound probability density function, such as the lognormal distribution, with a mean determined by a multiplicative, autoregressive lag-1 formulation typical of chain-ladder models, i.e. $p(y_{ij} \mid y_{ij-1}, \alpha_{j-1}, \sigma) = \text{Lognormal}(y_{ij} \mid \log(\alpha_{j-1} y_{ij-1}), \sigma)$.

The choice of prior distribution is, historically, one of the most contended aspects of Bayesian modeling, and one must justify why priors were selected (Winkler, 1967; Kruschke, 2021;

Mikkola et al., 2023). We support, as a starting point, the use of *weakly-informative* prior distributions that are sufficiently diffuse to allow a broad range of parameter values without reducing model performance (e.g. resulting in underflow or overflow of numerical computations) or realism. If independent expert knowledge is available, prior elicitation from multiple domain experts may be used, or from previous data if independent datasets are available (Falconer et al., 2022; Mikkola et al., 2023). Nonetheless, weakly informative prior distributions are still a useful starting point for model development because they appeal to the principles of starting simply and failing fast (Gelman et al., 2020), from which more informative priors can be developed. It is often wise to scale the data, or choose model parameterisations, in such a way that a standard normal distributions, or other generic prior distributions, would satisfy the definition of weakly informative. For instance, for the chain-ladder model above, a standard normal prior on the log-scale link ratios, $\log \alpha$, would imply a median link ratio of 1.0 on the lognormal scale, with a mean of 1.7 and a standard deviation of 2.2, which is suitably realistic yet diffuse enough to capture a range of average loss development dynamics. For σ, the residual standard deviation, we might first choose to scale the losses by its standard deviation, or some large value, to ensure that a standard normal prior distribution on $\log \sigma$ is also appropriate. These priors are easily generalizable to different datasets if the same data scaling is applied.

Together, a variant of the chain-ladder model can be written mathematically as:

$$\begin{aligned} y_{ij} &\sim \text{Lognormal}(\mu_{ij}, \sigma) \\ \mu_{ij} &= \log(\alpha_{j-1} y_{ij-1}) \\ \log \alpha &\sim \text{Normal}(0, 1) \\ \log \sigma &\sim \text{Normal}(0, 1) \end{aligned} \tag{1}$$

A useful exploratory method of checking model assumptions during model development is prior predictive checks (Gabry et al., 2019; Kruschke, 2021; Gelman et al., 2020). Prior predictive checks simulate data from the full generative model without fitting the model, which are samples from the prior predictive distribution, allowing modelers to compare the simulated data to *a priori* expectations and expert knowledge. Prior predictive checks are particularly useful when working with complex models where the implications of a set of parameters on resulting losses is unintuitive or otherwise difficult to reason about, as is the case with most models that we use regularly. For example, wide priors on the chain-ladder link ratios (α) can create an "explosion" or "crash" in loss predictions across development if they are too high or too low, leading to numerical overflow and underflow, respectively. Ideally, priors can be selected to ensure that losses roughly grow and eventually plateau in each accident period, leaving enough room for variable types of dynamics, but not allowing for degenerate model predictions. We leave specific examples of prior predictive checks for Section 3.4.2.

In this paper, we fit all Bayesian models using Stan (Carpenter et al., 2017), via the Python interface `CmdStanPy` (Stan Development Team, 2024b) and command line interface `CmdStan`

(Stan Development Team, 2024a). Stan uses Hamiltonion Monte Carlo, a variant of Markov chain Monte Carlo, to sample from the posterior distribution, and returns a number of diagnostic criteria for modelers to inspect the convergence of their Markov chains. We do not go into the details of convergence diagnostics for Markov chain Monte Carlo here, as these have been covered in detail elsewhere (e.g. Gelman et al., 2013). Our Bayesian workflow is agnostic to software or implementation choice, and so our discussion below is not particular to Stan.

2.2 Model validation

Validation of Bayesian models ideally starts before data is collected. While prior predictive checks are one type of model validation, a more principled approach fits the model to multiple sets of data simulated from the prior predictive distribution to inspect the accuracy and calibration of parameter recovery. The now dominant paradigm for the latter task is *simulation-based calibration* (Talts et al., 2018; Modrák et al., 2023), which is motivated by the *self-consistency* property of Bayesian models. Put simply, fitting the model to many instances of data simulated from the prior predictive distribution, and averaging across each resulting posterior distribution, should return the prior distributions for each parameter, or other quantity of interest. Taking the first link ratio in the Bayesian chain-ladder example, α_1, if the model is working as intended, we would expect the posterior means of $\log \alpha_1^{\prime(m)}$, where the superscript indicates a posterior distribution (the prime symbol) from the mth prior predictive sample, to match the standard normal prior distribution, $p(\log \alpha) = \text{Normal}(0, 1)$. It is important to also inspect the distribution of test quantities that fold in the complete data space, such as the joint log likelihood (Modrák et al., 2023). Typically, the posteriors are summarized not by their means but using rank order statistics (Casella and Berger, 2002), $R(\theta) = \sum_{s=1}^{S} \theta > \theta^{\prime(s)}$, where θ represents any parameter or generated quantity from the prior predictive distribution, and $\theta^{\prime(s)}$ is the sth sample from the estimated posterior distribution. If the model is working as intended, then a histogram of $R(\theta)$ should be approximately uniformly distributed, for which graphical and numerical tests of uniformity can be used to confirm. Departures from uniformity indicate biases in parameter estimation (skewed histograms), estimates that are too certain (U-shaped histograms), or estimates that are too uncertain (inverted U-shaped histograms).

Simulation-based calibration additionally provides the modeler access to a large number model fits, from which other useful model validation quantities can be inspected. For instance, the proportion of time a parameter θ falls within some interval of the estimated posterior distribution θ' across simulated datasets informs the modeler about parameter calibration, and the percentile of the real observations on the posterior predictive distribution informs us about the calibration of the model forecasts (Gneiting et al., 2007). Moreover, convergence diagnostics for the Markov chain Monte Carlo samplers can be aggregated and any systematic computational problems identified.

In addition to simulation-based calibration, model validation can make use of posterior

predictive checks. Like prior predictive checks, posterior predictive checks can be both numerical or purely graphical inspections of model fit to the data (Guttman, 1967; Rubin, 1984; Gelman et al., 1996), most usefully the actual data, or types of data, that the model will be fit to in its real-world applications. Posterior predictive checks allow for us to determine if a model is able to capture qualitiative features in the observed data that align with expert opinion or intuition. For example, loss development models should be specified such that the uncertainty in predicted losses levels off as development lag increases. Leveling off is expected due to the fundamentals of how insurance claims are paid off over time until there are none left to pay, leading to asymptotic behavior of losses across development lags within accident years. If posterior predictions from the chain-ladder model fitted to real-world data show that losses grow without bound across development, it is likely that the model is improperly specified with respect to the data-generating process in some way.

An additional benefit of posterior predictive checks is that they can be used to compute other quantities of interest, including both in- and out-of-sample fit statistics that can be used for model comparison. In our work, we often inspect calibration of the posterior predictive distribution by calculating the proportion of true values that fall within some interval width of the posterior samples, or compute some quantity of distance of the prediction from the true values (see e.g. Rubin, 1984; Gneiting et al., 2007; Gelman et al., 2020, for futher examples). Such metrics can be used to gain a high-level understanding of model performance, including if the model makes predictions that are biased or over/under-confident in some way.

2.3 Model comparison, selection and averaging

The final main component of the Bayesian workflow we cover here is how to approach the comparison of competing models, and the eventual selection or averaging of competing models. Model comparison, selection, and averaging is an expansive topic, from the choice of quantites to use to score models (e.g. Gneiting and Raftery, 2007; Vehtari and Ojanen, 2012; Piironen and Vehtari, 2017), to the method of model averaging (e.g. Carlin and Chib, 1995; Hoeting et al., 1999; Yao et al., 2018, 2022). Below, we follow modern Bayesian model comparisons by using the expected log pointwise predictive density (ELPD) (Vehtari et al., 2017) to score models, which is based on the logarithmic score (Good, 1952) and defined as:

$$
\begin{aligned}
\text{ELPD} &= \log \int p(\tilde{y} \mid y) d\theta \\
&= \log \int p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta \\
&\approx \sum_{i=1}^{N} \frac{1}{S} \log p(\tilde{y}_i \mid \theta^s) p(\theta^s \mid y) \\
&\approx \sum_{i=1}^{N} \text{ELPD}_i
\end{aligned}
\tag{2}
$$

where $\tilde{y}$ is external, out-of-sample data not used during model fitting, and the approximation in the last line reflects the Monte Carlo approximation of ELPD from posterior samples. Conceptually, ELPD is a measure of the height of the posterior predictive density at the true value for each datapoint, aggregated across all out-of-sample datapoints. We focus on out-of-sample predictive ablity when comparing models, rather than in-sample measures of model fit. The latter is useful for posterior predictive checks, and model development cycles, whereas the former matches the goals of predictive inference, and securitizing casualty insurance risks, more closely. When external data to score models are not available, approximate methods may be employed, namely cross validation, such as approximate leave-one-out cross-validation via Pareto smoothed importance sampling (Vehtari et al., 2017). For two models, a and b, the difference between ELPDs is often of interest, $\text{ELPD}_{\text{diff}} = \text{ELPD}_a - \text{ELPD}_b$, and potential measures of uncertainty such as the standard error of the differences (Vehtari et al., 2017; Sivula et al., 2023). Other scoring rules are of course possible, such as quadratic scoring rules based on the squared errors of the predictions from the true observations (Selten, 1998), and scoring rules based on the predictive cumulative distribution function (Gneiting and Raftery, 2007; Gneiting et al., 2007).

While model selection takes a "winner takes all approach", choosing the model that has the best score (e.g. the highest ELPD), model averaging blends model predictions with weights proportional to their predictive performance. The canonical Bayesian paradigm is Bayesian model averaging (e.g. Hoeting et al., 1999), which uses Bayes' rule to estimate the posterior probablity of model k given the data, $p(M_k \mid y)$, and uses those posterior model probabilities as weights to blend model predictions. Bayesian model averaging only considers the in-sample performance, however, and, in the limit of infinite data, will assign the model with the best in-sample fit 100% weight (Yao et al., 2018; Haines and Goold, 2024). Consequently, methods that estimate model weights using measures of out-of-sample model performance are generally favoured, reducing the risk of generalization error to future data. Of the various approaches, model stacking (Yao et al., 2018) provides a principled method of estimating model weights for Bayesian methods, which is highly extensible (Yao et al., 2022; Haines and Goold, 2024). Stacking finds model weights that maximize the average expected log pointwise predictive densities, ELPD_i, across data points.

3 A Bayesian workflow for casualty ILS

With the basics of a Bayesian workflow now outlined, we now turn attention to the Bayesian workflow that we follow when securitizing casualty insurnace risk. To start, we emphasize that securitization of insurance programs involves two separate stages–the loss modeling stage and the deal modeling stage. The goal of loss modeling is to estimate ultimate losses for future accident years, including how long such ultimate losses take to develop. Outputs from loss modeling then serve as input to deal models, which are used estimate underwriting and investment cashflows given idiosyncratic deal terms that are negotiated between investors and insurers. Due to the idiosyncratic nature of deal modeling, our focus here is primarily on the loss modeling stage. However, due to how important deal modeling is for securitization, we include a brief overview at the end to describe how to get from loss forecasts to the underwriting and investment cashflows that make up an ILS deal.

Figure 1 provides a visual representation of our Bayesian workflow. In the sections that follow, we walk through each step of the workflow in detail. Below, we start by defining generative models for loss development and forecasting.

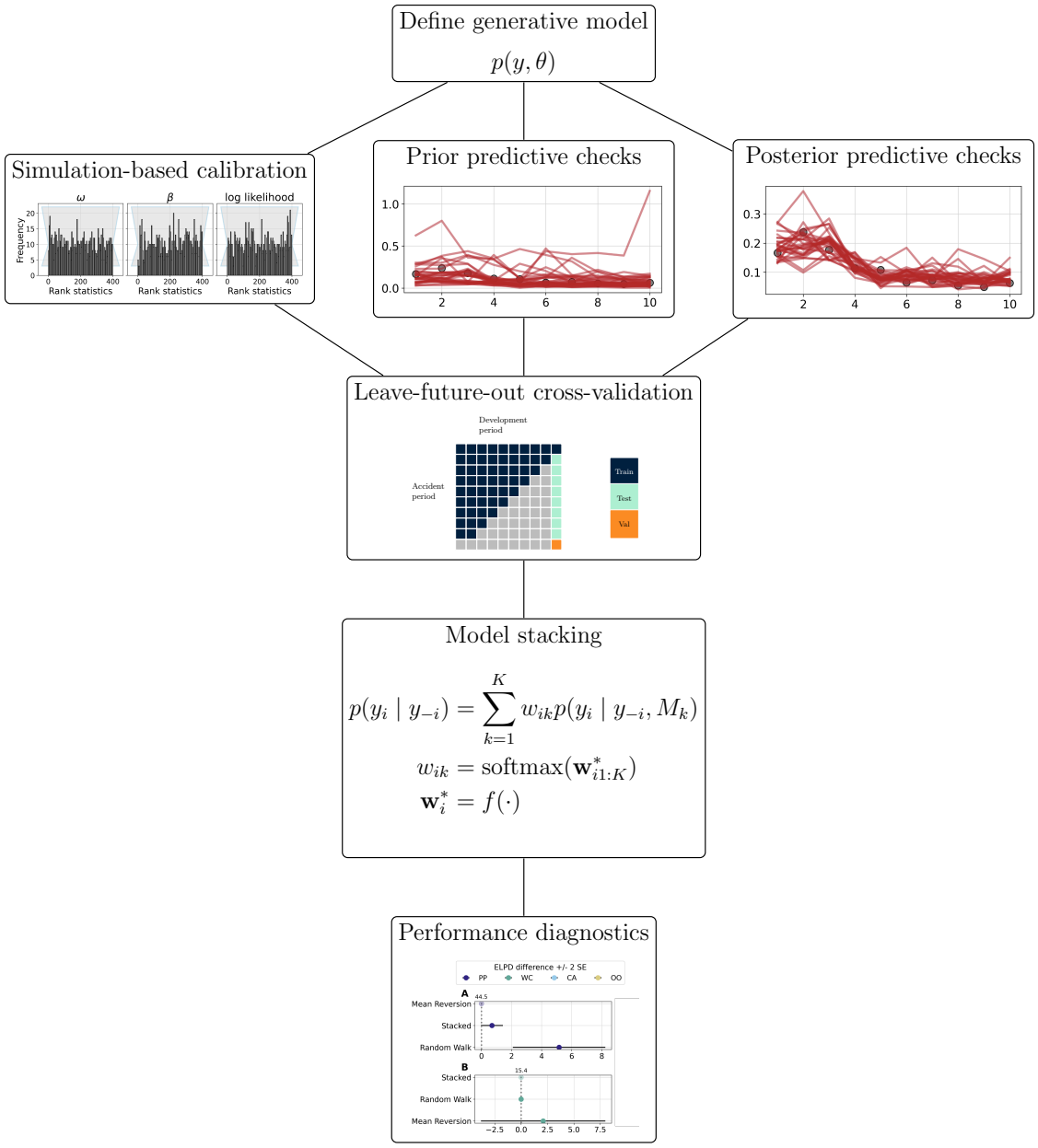


Figure 1: A Bayesian workflow for modeling casualty insurance loss development and forecasting future losses – the first step for securitization. In-depth explanations for each step are described throughout Section 3. Here, θ represents all parameters in the model, and $p(y, \theta)$ then represents the joint distribution of data and model parameters. Graphs in the workflow steps here are only included for illustration; we dive into each workflow step in more detail in the following sections.

3.1 Historical loss development

There is an extensive body of literature on loss development in the actuarial literature, ranging from link-ratio models that directly capture multiplicative changes in cumulative losses between development lags (Mack, 1994; England and Verrall, 2002) to parametric growth or decay models that capture cumulative losses directly (Zhang et al., 2012). Link-ratio models are useful for capturing volatile patterns of change often observed across early development lags (which we refer to as the "body" of the loss triangle), but they are less identifiable for later development lags (commonly referred to as the "tail" of the loss triangle) due to the lack of observed data. Poor identifiability in the tail is problematic given our goal to extrapolate out to the ultimate loss for each accident year. Conversely, parametric models are better-suited for extrapolation, but they tend to be less flexible in the body.

To resolve issues with using link-ratio versus parametric models in isolation, we follow CAS Tail Factor Working Party (2013), using variants of the chain-ladder and generalized Bondy models for the body and tail processes, respectively (Mack, 1993; CAS Tail Factor Working Party, 2013).

Following the notation of Goold (2024), we use $\mathcal{Y}$ to denote the loss development triangle for an aggregated pool of insurance policies, defined by:

$$\mathcal{Y} = \{y_{ij} : i = 1, ..., N; j = 1, ..., N - i + 1\} \tag{3}$$

where $i = 1, ..., N$ indicates the accident year and $j = 1, ..., M$ indicates the development lags. In real-world data, losses for a given accident year i are only known up to development lag $j = N - i + 1$, creating the triangular data structure that loss triangles are named for. Loss development models seek to predict the complement of $\mathcal{Y}$, or $\tilde{\mathcal{Y}}$, where the goal is to estimate $\tilde{y}_{i\infty}$ $\forall i = 1, ..., N$.

3.1.1 Body development

As described in Section 2.1, the chain-ladder model is a useful starting point for defining a loss development model more generally. However, the base model presented earlier can be extended to better capture how uncertainty in ultimate losses changes as a function of development lag. Further, it needs modification to capture the body versus tail distinction we make above. Our fully-specified Bayesian variant of the chain-ladder model is defined as follows:

$$
\begin{aligned}
y_{ij} &\sim \text{Lognormal}(\mu_{\text{ij}}, \sigma_{\text{ij}}) \\
\mu_{ij} &= \log(\alpha_{j-1} y_{ij-1}) \\
\sigma_{ij}^2 &= \exp(\gamma_1 + \gamma_2 j + \ln(y_{ij-1})), \quad \forall j \in (1, \tau] \\
\log \boldsymbol{\alpha}_{1:M-1} &\sim \text{Normal}(0, 1) \\
\gamma_1 &\sim \text{Normal}(-3, .25) \\
\gamma_2 &\sim \text{Normal}(-1, .1)
\end{aligned}
\tag{4}
$$

where $\boldsymbol{\alpha}$ is the vector of "development factors" that capture how losses change across development lags. The prior on $\boldsymbol{\alpha}$ implies a median age-to-age factor of 1.0 across development lags, with a lower 5%ile of ≈ 0.2 and upper 95%ile of ≈ 5. Therefore, the prior is quite diffuse, allowing for a wide variety of development patterns that we might expect to see in real data. We purposefully choose diffuse priors over flat priors on age-to-age factors because the latter can often produce estimates that are completely unrealistic as determined by prior predictive checks (see Section 3.4.2).

In most ARIMA or state-space style time-series models, we expect that uncertainty would increase across time – for loss development models, uncertainty propagation is a bit more complex. Intuitively, the further into the future we are when looking back at losses for a given accident year, the more claims are closed, and as a result we are more certain in the ultimate loss. To capture this behavior, we assume that variance in the ultimate loss follows a linear model with an intercept and slope captured by $\boldsymbol{\gamma} = (\gamma_1, \gamma_2)'$. We expect $\gamma_2 < 0$, indicating that losses vary around μ_{ij} less as development progresses. This decrease in variability is critical to capture the asymptotic behavior of losses across development – without this heterogeneity mechanism, the model predictions will be too certain for early development lags and too uncertain for later development lags.

However, it is important to distinguish between observation noise (or σ_{ij}^2, the variance around μ_{ij}) and parameter uncertainty (uncertainty in $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$). Observation noise arises from epistemic uncertainty in loss development, whereas parameter uncertainty stems from our limited knowledge of the true values of $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$. Because each μ_{ij} is a function of α_{j-1} and the previous observation y_{ij-1} (or $\hat{y}_{ij-1}$ if the previous loss is itself a prediction), uncertainty compounds across development. This compounding effect is particularly clear when using the model for "forecasting" beyond observed development years.

Due to the two-stage nature of our loss development workflow, the chain-ladder model is only fitted to losses where $j \leq \tau$, where $\tau \in 2, ..., M$ is an integer chosen by an analyst based on their knowledge of how long the body development process should be active. Generally, τ should be lower if the program is quick to develop (reaching the tail process for lower development lags), and higher if the program is slow to develop.

Finally, it is worth noting that the lognormal specification above could be replaced by a number of other distributions, including gamma, inverse-gamma, ex-Gaussian, and more. If frequency data is available, a frequency-severity model could be used to parse out different

components of the underlying losses. We use the lognormal distribution here for simplicity, as our main goal is to illustrate how models from various stages of the workflow fit together.

3.1.2 Tail development

Next, for the "tail" model, we follow from the actuarial literature (CAS Tail Factor Working Party, 2013), using a model based on the Bondy method. Our Bayesian "generalized Bondy" model is specified as follows:

$$
\begin{aligned}
y_{ij} &\sim \text{Lognormal}(\mu_{\text{ij}}, \sigma_{\text{ij}}) \\
\mu_{ij} &= \log(\alpha_{ij} y_{ij-1}) \\
\alpha_{ij} &= \omega^{\beta^j} \\
\sigma_{ij}^2 &= \exp(\lambda_1 + \lambda_2 j + \ln(y_{ij-1})), \quad \forall j \in [\rho_1, \rho_2] \\
\log \omega &\sim \text{Normal}^{+}(0, 1) \\
\log \frac{\beta}{1-\beta} &\sim \text{Normal}(-2, .5) \\
\lambda_1 &\sim \text{Normal}(-3, .25) \\
\lambda_2 &\sim \text{Normal}(-1, .1)
\end{aligned}
\tag{5}
$$

where the model form is identical to the chain-ladder model defined above, except the link ratios $\boldsymbol{\alpha}$ are now determined by a parametric decay model as opposed to being unconstrained free parameters. In the decay model, the asymptote ω is constrained to have a lower bound of 1.0 due to the truncated normal prior, which captures our assumption that losses will themselves asymptote to some ultimate value as develoment progresses. β then controls the speed of decay, which we expect to be higher for more "long-tailed" lines of business (e.g., worker's compensation). However, values for β too close to 1.0 can produce degenerate behavior where losses only asymptote at extreme development lags. Therefore, we center the prior such that decay in the resulting link ratios matches the timescale we expect (i.e., years-to decades-long decay). Unlike the chain-ladder model, the generalized Bondy model is fitted to only the window of development lags $j \in [\rho_1, \rho_2]$, where $(\rho_1, \rho_2) \in 2, ..., M$, $\rho_1 < \rho_2$, are chosen by an analyst based on where the tail process is assumed to begin and end.

After fitting both the chain-ladder and generalized Bondy models, predictions are made by forward simulation starting from the left edge of the loss triangle for each accident year (i.e. $y_{i,1}$). The chain-ladder model is used to generate posterior predictions up to only $j = \tau$, and the generalized Bondy model is then used to generate posterior predictions from for all $j > \tau$ out to a development lag j that is sufficiently large as to be practically indiscernable from $j = \infty$. These posterior predictive distributions for $\tilde{y}_{i,\infty} \forall i = 1, ..., N$ are then our "developed ultimate losses".

3.2 Future ultimate loss forecasting

Loss development models produce predictions for the ultimate losses associated with each historic accident year (or experience period more generally), yet the underlying risk being securitized is based on the future performance of a pool of policies yet to be written. Typically, securitization involves the ultimate losses in the next one to three accident years, or $\tilde{y}_{i,\infty} \forall i = (N+1, N+2, N+3)$. To solve this problem, we use the developed ultimate losses from the loss development model (i.e. $\tilde{y}_{i,\infty} \forall i = 1, ..., N$) as input to a time-series model used to forecast ultimate losses for future accident years. Bayesian models are especially useful for this step, as they allow for us to easily incorporate uncertainty in the loss development model posterior predictions into the forecasting model.

The state-space modeling framework is particularly well-suited for loss forecasting, as it allows for us to develop models that are informed by expert intuition regarding how losses evolve over time. In practice, we use a variety of different forecasting models dependent on use-case. For this example, we will overview a state-space random walk and state-space mean reversion model that illustrate our basic workflow.

Before defining the models, we will introduce new notation. Although we find it useful for loss development to be directly on the losses, we find it easier to work with loss ratios for the forecasting step. Below, we use r to indicate the loss ratio, such that $r_{ij} = \frac{y_{ij}}{p_i} \ \forall i = 1, ..., N, \forall j = 1, ..., M$, where p_i is the premium volume for accident year i.

3.2.1 A basic random walk model

The base random walk model is defined as the following state-space model (Hyndman and Athanasopoulos, 2018):

$$
\begin{aligned}
r_{i\infty} &\sim \text{Lognormal}(\eta_i, \sigma_i) \\
\eta_i &\sim \begin{cases} \text{Normal}(\eta_0, \epsilon), & \text{if } i = 1 \\ \text{Normal}(\eta_{i-1}, \epsilon), & \text{otherwise} \end{cases} \\
\sigma_i^2 &= \exp(\gamma_1)^2 + \exp(\gamma_2)^2 / \sqrt{p_i} \\
\log \epsilon &\sim \text{Normal}(-0.5, 1) \\
\eta_0 &\sim \text{Normal}(0, 1) \\
\boldsymbol{\gamma}_{1:2} &\sim \text{Normal}(-2, 1)
\end{aligned}
\tag{6}
$$

where $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_N)'$ is a latent (in log-scale) ultimate loss ratio that drifts across accident years with deviations between years proportional to ϵ. The "observed" ultimate loss ratio ($r_{i\infty}$) then follows a lognormal distribution with observation noise σ_i that is a function of earned premium for the given accident year (p_i) such that as earned premium increases, observation noise decreases.

3.2.2 Adding mean reversion

The base state-space model is appealing because it allows us to separate observation noise from the "true" latent variability in the underlying ultimate loss ratios. The distinction is important, particularly in cases where the premium volume of a program changes significantly over time, or when extending the model to capture effects that influence the latent (but not observation) process. One example that comes up frequently in our own work is the addition of a mean-reverting mechanism to the random walk process:

$$\begin{aligned} \eta_i &\sim \begin{cases} \text{Normal}(\mu(1-\phi) + \eta_0\phi, \epsilon), & \text{if } i = 1 \\ \text{Normal}(\mu(1-\phi) + \eta_{i-1}\phi, \epsilon), & \text{otherwise} \end{cases} \\ \mu &\sim \text{Normal}(-1, 1) \\ \log\frac{\phi}{1-\phi} &\sim \text{Normal}(0, 1) \end{aligned} \tag{7}$$

where μ is the latent mean that the random walk process reverts back to, and ϕ indicates the strength and direction of the reversion process. All other model terms are identical to the random walk model defined above. Note that the mean reversion process is only on the latent log loss ratio scale, and the observation equation is unchanged. The motivation for mean reversion on the latent loss ratios is straightforward – the underlying insurance program is being managed by a team that is constantly updating their expectations regarding how loss ratios will develop for the associated policies. The management team can then influence how future loss ratios unfold by changing underwriting practices, increasing or decreasing premium rates, and more. The end result is that ultimate loss ratios tend toward some target value as management makes decisions to ensure that the program is successful.

3.2.3 Adding measurement error

The forecasting models above are defined with $r_{i\infty}$ as the target. However, in real-world scenarios, we do not know $r_{i\infty}$ – instead, we have the posterior predictions from the loss development models, or $\tilde{y}_{i\infty}$ (transformed to the loss ratio scale per $\tilde{r}_{i\infty} = \frac{\tilde{y}_{i\infty}}{p_i}\ \forall i = 1, ..., N$). To account for uncertainty in these posterior predictions in the context of the forecasting model, we can add an error-in-variables measurement error model (Stefanski, 2000) to the forecasting models that takes the posterior means $\mathbb{E}\left[\tilde{r}_{i\infty}\right]\ \forall i = 1, ..., N$ and posterior standard deviations $\mathbb{SD}\left[\tilde{r}_{i\infty}\right]\ \forall i = 1, ..., N$ as input and models the "true" underlying ultimate loss ratio $r'_{i\infty}$. For both the random walk and mean reversion models, this involves the following modification to the likelihood expression:

$$
\begin{aligned}
{r'}_{i\infty} &\sim \text{Lognormal}(\eta_i, \sigma_i) \\
\mathbb{E}\left[\tilde{r}_{i\infty}\right] &\sim \text{Lognormal}(\mu_\xi, \sigma_\xi) \\
\mu_\xi &= \log \frac{{r'}_{i\infty}^2}{\sqrt{{r'}_{i\infty}^2 + \mathbb{SD}\left[\tilde{r}_{i\infty}\right]^2}} \\
\sigma_\xi &= \sqrt{\log\left(1 + \frac{\mathbb{SD}\left[\tilde{r}_{i\infty}\right]^2}{\mathbb{E}\left[\tilde{r}_{i\infty}\right]^2}\right)}
\end{aligned}
\tag{8}
$$

Here, μ_ξ and σ_ξ are the mean and standard deviation terms capturing the relationship between the true (log) ultimate loss ratios ${r'}_{i\infty}$ and the observed posterior means from the development model posterior predictions. The true ultimate loss ratios are then used in the likelihood expression as the target for forecasting.

Note that ${r'}_{i\infty}$ in equation 8 is now a model parameter with its own prior. Above, the lack of prior implies a uniform distribution, which implies that the true ultimate loss ratio could take on any value. Of course, we often have quite a bit of information on what the distribution of ultimate loss ratios should be. Given the empirical mean ($\mathbb{E}\left[r\right]$) and standard deviation ($\mathbb{SD}\left[r\right]$) of ultimate loss ratios that we have observed in historic data, we can specify a prior on ${r'}_{i\infty}$ that helps constrain the model to realistic values (see Section 3.5 for an example):

$$
\begin{aligned}
{r'}_{i\infty} &\sim \text{Lognormal}(\mu_{{r'}_{i\infty}}, \sigma_{{r'}_{i\infty}}) \\
\mu_{{r'}_{i\infty}} &= \log \frac{\mathbb{E}\left[r\right]^2}{\sqrt{\mathbb{E}\left[r\right]^2 + \mathbb{SD}\left[r\right]^2}} \\
\sigma_{{r'}_{i\infty}} &= \sqrt{\log\left(1 + \frac{\mathbb{SD}\left[r\right]^2}{\mathbb{E}\left[r\right]^2}\right)}
\end{aligned}
\tag{9}
$$

where $\mu_{{r'}_{i\infty}}$ and $\sigma_{{r'}_{i\infty}}$ are akin to μ_ξ and σ_ξ in equation 8, representing the mean and standard deviation of the historic (log) ultimate loss ratios used to derive the informed prior. We find that such data-informed priors can often help with model diagnostics, including both convergence of the sampler and out-of-sample model performance metrics such as RMSE and ELPD.

In practice, there are many potential measurement error assumptions one can make, and the above is just one example. In the current context, lognormal measurement error is reasonable given that the development model predictions are generated from a lognormal distribution. An alternative approach would be to jointly model the loss development and forecasting stages such that there is no need for an intermediate summarization step and subsequent measurement error modeling (akin to Goold (2024), who jointly models the body and tail

stages of loss development). However, in practice we find that it is useful to separate the loss development from forecasting stages, which allows for more intuitive setting of priors on the individual models and easier investigation of model behavior. Further, having separate stages keeps the flow of information forward such that the forecasting model does not influence development model parameter estimation.

3.3 Deriving data-driven priors

When defining models in previous sections, we only made conceptual arguments for why particular priors were chosen. Priors can have quite a big impact on ultimate loss forecasts, so it is important to choose them in a data-driven way. This is particularly true if the triangle has minimal history, as is often the case for new programs undergoing ILS vetting. For example, say we are assessing a commercial auto program for a securitization opportunity, and the program has only existed for the past 5 years. With only 5 years of development, we may not have enough data to fit the tail portion of the loss development model (see Section 3.1.2). Even if we were able to fit an adequate tail model, we would be left with only 5 accident years worth of ultimate loss ratios to feed into the rather complex forecasting models in Section 3.2. In such cases, relying on models with uninformative or weakly informative priors can produce ultimate loss ratio predictions with uncertainty intervals far too wide to make ILS viable. Conversely, if we set priors that are too informative to combat low-data scenarios, we run the risk of biasing our model parameters, resulting in inaccurate and overconfident predictions that make an ILS deal appear more (or less) appealing than it is in reality.

Our solution to this problem is to use hierarchical Bayesian analysis (Gelman, 2006) to derive a set of data-driven priors for each model and line of business that we commonly work with. Hierarchical Bayesian analysis involves specifying a hierarchical model where the model parameters for each individual loss triangle themselves follow a group-level distribution. The intuition behind hierarchical models is simple – in the absence of any historical loss information for a given program, our best guess for the program's development pattern would be the development pattern for similar programs. Hierarchical models allow for us to estimate such group-level distributions from a pool of programs, which we can then use as priors for new programs undergoing ILS vetting.

The rationale for hierarchical models extends beyond intuition. Hierarchical models are closely related to credibility theory, which has informed actuarial practice for over a century (Michelbacher, 1918). In particular, the mathematical form of hierarchical pooling is identical to a common form of classical credibility weighting (c.f. equations 2 and 3 in Gelman and Pardoe (2006) to equations 25 and 24 in Whitney (1918), respectively). In both cases, the relative proportion of within- versus across-program variance is used to pool a program-level estimate toward the group-level average such that programs with higher volatility are more strongly pooled toward the group average and *vice versa*.

As an example, the hierarchical variant of the random walk model (excluding measurement

error) is as follows:

$$
\begin{aligned}
r_{gi\infty} &\sim \text{Lognormal}(\eta_{gi}, \sigma_{gi}) \\
\eta_{gi} &\sim \begin{cases} \text{Normal}(\eta_{g0}, \epsilon_g), & \text{if } i = 1 \\ \text{Normal}(\eta_{gi-1}, \epsilon_g), & \text{otherwise} \end{cases} \\
\sigma_{gi}^2 &= \exp(\gamma_{g1})^2 + \exp(\gamma_{g2})^2 / \sqrt{p_{gi}} \\
\log \epsilon_g &\sim \text{Normal}(\epsilon_\mu, \epsilon_\sigma) \\
\eta_{g0} &\sim \text{Normal}(\eta_{\mu 0}, \eta_{\sigma 0}) \\
\boldsymbol{\gamma}_{g(1:2)} &\sim \text{Normal}(-2, 1) \\
\epsilon_\mu &\sim \text{Normal}(-2, 0.5) \\
\log \epsilon_\sigma &\sim \text{Normal}(-2, 0.5) \\
\eta_{\mu 0} &\sim \text{Normal}(-1, 0.5) \\
\log \eta_{\sigma 0} &\sim \text{Normal}(-2, 0.5)
\end{aligned}
\tag{10}
$$

where $g = 1, 2, ..., G$ is the program index within a group of similar programs. The model is functionally equivalent at the individual loss triangle level, but now instead of having hard-coded priors on the triangle-level parameters, the group-level parameters (or hyper-parameters) that are estimated from the data. For example, $\log \epsilon_g$ now follows a group-level normal distribution with log mean and log standard deviations ϵ_μ and ϵ_σ, respectively. This scheme does require specifying priors on the group-level parameters, but their influence is rather minimal if the number of programs G is sufficiently high enough to allow for precise estimation of the group-level parameters. We are often in this situation, where we have many example triangles for each line of business we work with, yet each triangle has minimal observations individually. The hierarchical model allows for partial pooling of information across triangles such that information on parameters for each triangle informs the group-level parameter, in turn informing all other triangle-level parameters.

In practice, it is not computationally efficient to fit a hierarchical model to all historic triangles in addition to the triangle of interest when we are analyzing data for a new program. Therefore, we often hierarchical models in a two-step fashion. First, we define the grouping variable of interest, which determines which triangles will be included in a given hierarchical model. Typically, we group triangles by line of business (e.g., whether a triangle is from a commerical auto, private passenger, general liability, or worker's compensation insurance program). Next, we limit our focus to only recent triangles (e.g., if it is 2024, we may only consider triangles from 2014 onward). Once the group is defined, we fit the hierarchical model to the data within the group (and undergo model validation exercises described below in Section 3.4) and then summarize the group-level parameters. For example, once we have posteriors from the hierarchical model for ϵ_μ and ϵ_σ, the prior on the non-hierarchical model (equation 6) becomes $\log \epsilon \sim \text{Normal}(\mathbb{E}[\epsilon_\mu], \mathbb{E}[\epsilon_\sigma])$. When analyzing a new program that fits into one of the pre-defined lines of business, we use the corresponding group-level parameter summaries as priors in the non-hierarchical variant of the model, thus obtaining

the hierarchical pooling (or credibility weighting) described above in a dynamic, data-driven way. Note that this approach assumes that we can estimate the group-level parameters with sufficient precision, which is often the case when we have many programs within a given line of business. In cases where we have fewer programs within a group, it is sensible to scale the empirical group-level standard deviation (i.e. $\mathbb{E}[\epsilon_\sigma]$) by some factor based on the number of programs in the group to ensure that the prior is not overly informative.

3.4 Model validation with simulated data

With the loss models defined, the next step is to validate the models using the suite of techniques from the Bayesian toolbox as described throughout Section 2.

3.4.1 Simulation-based calibration

Typically, we perform SBC for each stage of our modeling pipeline separately, for each new model we develop. For illustration, here we apply SBC to just the loss development modeling stage.

We simulated data from the loss development body and tail models using the exact priors specified when introducing the models above. We simulated 1000 separate datasets, each generated from a randomly sampled set of parameters from the afforementioned prior distributions. We set the body development lag cutoff to $\tau = 5$, and the tail development lag window to $\rho = (6, 10)$. Each dataset was generated to have 10 accident years and 10 development lags, thus matching the dimensions of a typical real-world loss triangle (the first accident year has 10 development lags of observed losses, second accident year has 9, third has 8, etc., until the 10th accient year has just 1).

For each of the 1000 simulated parameter sets and associated loss triangles, we fit the loss development model with $\tau = 5$ and $\rho = (6, 10)$, using the same set of priors used in the simulation. For each fit, we drew 1000 posterior samples for each of 4 indpendent MCMC chains. We then generated predictions out to the 10th development lag, thus filling out the lower diagonal of the triangle with posterior predictions. Next, we thinned the posterior samples such that we only kept overy 10th sample, resulting in a total of 400 posterior samples with negligible autocorrelation.

Finally, for each dataset and model fit pair, we computed the rank of the true parameter values and losses in the associated posterior distributions. Figure 2 shows the distribution of these ranks across all 1000 dataset and model fit pairs. Rank distributions for all parameters are indistinguishable from uniform distributions, indicating proper specification and good parameter recovery. A slight exception is for the posterior predictions, which show an inverted U shape. The inverted U shape signifies too much uncertainty in the posterior predictions. However, the ranks are almost entirely within the bounds we would expect for

data generated from a uniform distribution, so the extra uncertainty is minimal.

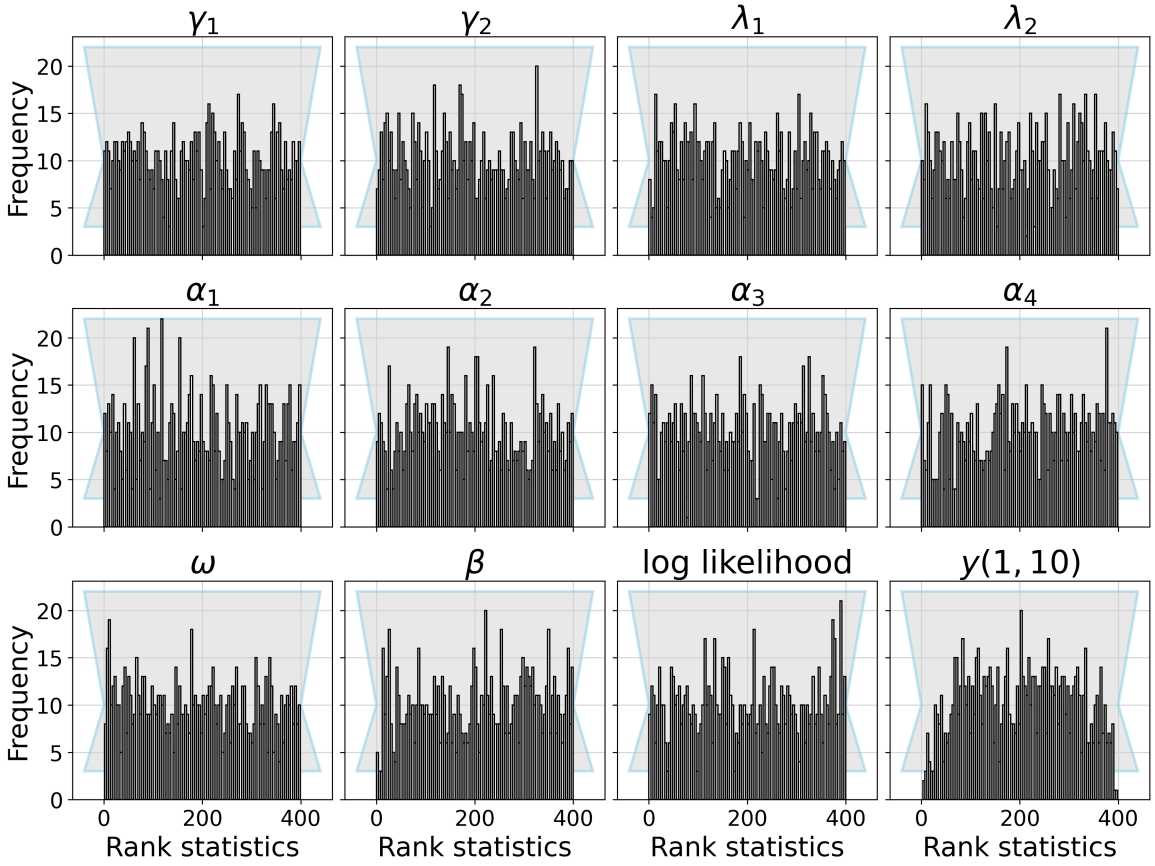


Figure 2: Simulation-based calibration rank histograms. For each model, we sampled 4000 draws from the posterior distribution, and thinned the samples by 10 to remove any autocorrelation, meaning a maxmimum rank statistic of 400. Uniformly distributed ranks indicate good calibration. Error ribbons indicate 99% uncertainty intervals for the target uniform distribution. Histogram bars falling outside of the intervals are then evidence of mis-calibration.

3.4.2 Prior and posterior predictive checks

Figure 3 illustrates prior predictive checks on three different example commercial auto programs from Meyers (2015). Note that we focus only on the forecasting model here for brevity, using the same priors defined in Section 3.2. To do so, we fit the loss development model to the triangle, then used the posterior predictions on the ultimates as input to the forecasting model as described in Section 3.2.3. The prior predictive simulations in Figure 3 are generally well-behaved, generating a wide range of loss patterns, most of which do not show the afforementioned "exploding" or"crashing" behaviors alluded to in Section 2.1. Compared to the real losses, the simulations tend to show much more variability, but this is expected before the model is conditioned on the real losses.

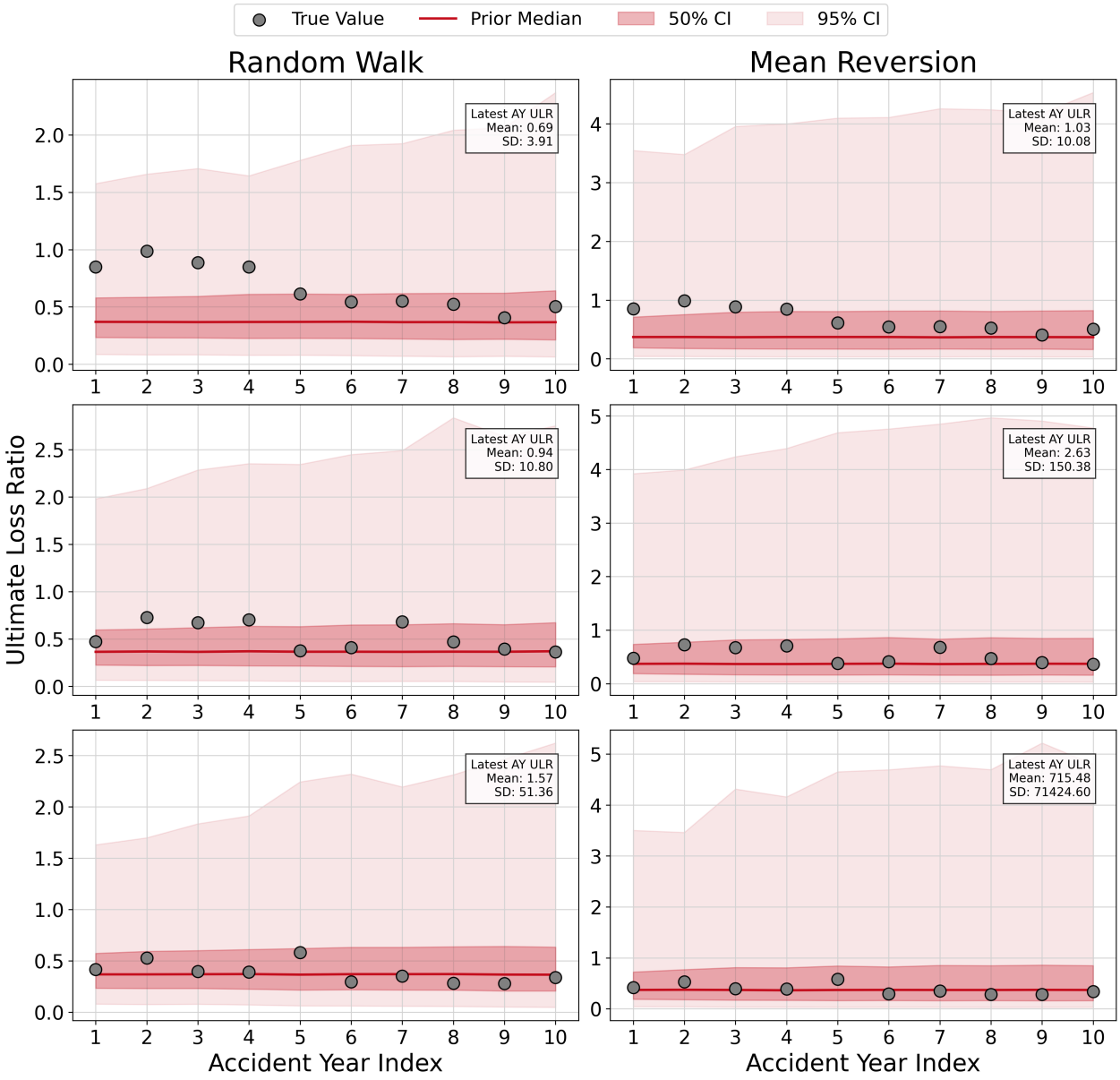


Figure 3: Prior predictions for three random commercial auto programs from the Meyers (2015) dataset. Shaded intervals indicate 50% and 95% credible intervals, and points are true losses at $y_{i(10)}$.

Figure 4 then shows posterior predictive checks for the same data in Figure 3, but after fitting the model to the real losses. The predictions now follow the real losses with much more precision, capturing the slow-moving drift in losses across accident years. Both the random walk and mean reversion models show a slight amount of over-estimation of losses for later accident years. However, some mis-esimtation of the true losses is expected given that the forecasting models are trained only on the development model predictions as input.

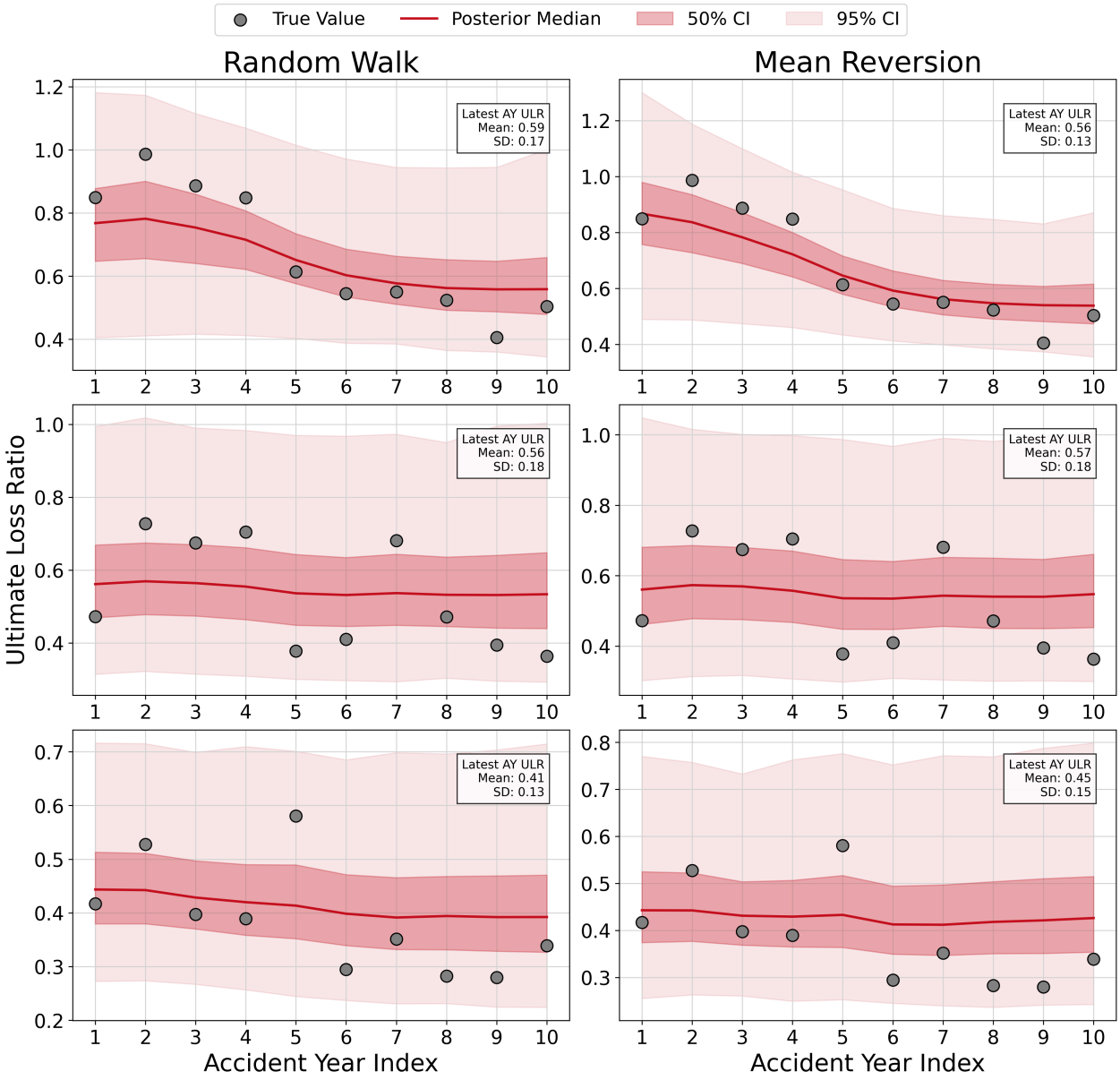


Figure 4: Posterior predictions for three random commercial auto programs from the Meyers (2015) dataset. The programs here are the same as in Figure 3. Shaded intervals indicate 50% and 95% credible intervals, and points are true losses at $y_{i(10)}$.

3.5 Backtesting workflow using empirical data

A model can pass all model validation checks but then fail to perform well on real-world prediction tasks given the wide range of different loss triangles that we encounter from day-to-day. Therefore, large-scale out-of-sample tests are crucial to determine if a given model (or model pipeline) is suited for production use. Further, we often have multiple competing models, where each model makes different assumptions regarding how losses change over

time. For example, equation 7 assumes mean reversion toward some latent ultimate loss, whereas equation 6 assumes no such latent ultimate loss or reversion process. An added benefit of conducting out-of-sample tests is that we can take advantage of modern Bayesian stacking methods that combine predictions from competing models, allowing for a combined performance better than the best fitting individual model. In the following sections, we overview our out-of-sample backtesting workflow and our use of stacking to improve model performance.

Often referred to as "backtesting", we use a form of exact leave-future-out cross validation (see Vehtari and Ojanen (2012) for a review of different cross-validation schemes). Backtesting in general is highly dependent on the specific data that one has available for use when testing. For our internal backtests, we license access to public statutory filings data via a vendor. This dataset contains information from insurance programs across over a dozen different lines of business, altogether comprising over 10,000 triangles, most with history dating back to the 1980's.

For demonstration purposes, here we will rely on an open dataset described by Meyers (2015), which contains about 50 loss triangles for each of four different lines of business, including private passenger auto, worker's compensation, commercial auto, and general liability lines. Each triangle in the Meyers dataset is 10 accident years by 10 development lags, which allows for us to fit our loss development to loss forecasting pipeline on each triangle exactly once, using the loss from the most recent accident year, development lag pair as the future, left-out datapoint for testing purposes (see Figure 5). Note that this scheme assumes $y_{i\infty} = y_{i(10)}\ \forall i = 1, ..., N$, which may not be true in practice for longer-tailed lines of business. However, given data constraints for this demonstration, we believe this is a reasonable assumption. We use $y_{i\infty}$ and $y_{i(10)}$ interchangeably in sections that follow.

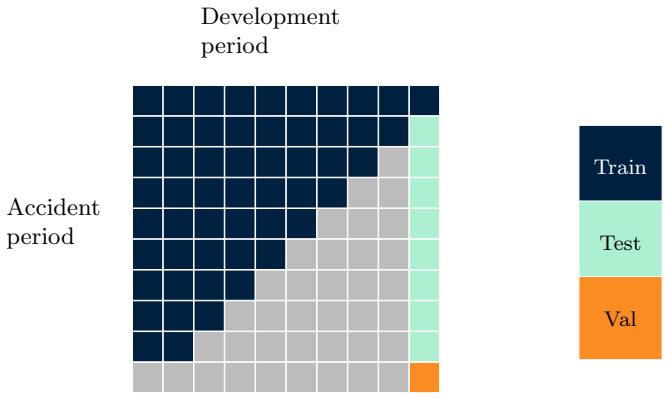


Figure 5: A schematic of the backtest procedure. Train datapoints are used to fit the loss development model, which is used to make ultimate loss predictions out to the right edge of the triangle. Predicted ultimate losses are then used as input for the forecasting models, which is used to generate out-of-sample predictions on both the Test and Validation datasets.

For the loss development stage, we fit the chain-ladder and generalized Bondy models to

each Meyers triangle indpendently using the specifications outlined in equations 4 and 5, respectively. For the shorter-tailed private passenger and commerical auto lines, we set the chain-ladder training threshold to $\tau = 4$, and the generalized Bondy training window to $\rho = (5, 10)$. For the remaining longer-tailed lines, we set $\tau = 6$ and $\rho = (4, 10)$. For all lines, we then generated posterior predictions out to $\tilde{y}_{gi(10)} \forall i = 1, ..., 9$ for each triangle g following the procedure in Section 3.1.2, which were used as ultimate losses for input to the forecasting models.

For the forecasting stage, we then fit both the random walk (equation 6) and mean reversion (equation 7) models hierarchically (see equation 10) across all triangles within each of the four lines of business in the Meyers dataset. Both models also used the measurement error specification in equation 8. We also used the measurement error prior defined in equation 9. Because $r_{g1(10)}$ is the only "observed" ultimate loss ratio for each triangle given our backtest scheme, we took the mean and standard deviation of these observed loss ratios across triangles within each line of business, and then used these in place of $\mathbb{E}[r]$ and $\mathbb{SD}[r]$, respectively, in equation 9. After fitting the models, we generated posterior predictions for each triangle for $\tilde{r}_{gi(10)} \forall i = 1, ..., 10$.

Finally, to determine how sensitive our results are to the prior specifications, we ran the full backtesting workflow pipeline with two alternative prior specifications – one with more diffuse priors, and one with more informative priors. We relegate details of this analysis to the appendix, but include some results below in the main text for completeness.

3.5.1 Stacking of predictive distributions

Once backtesting is complete, we can use the out-of-sample performance metrics to further improve predictive performance. In practice, we often use hierarchical Bayesian stacking to do so, a form of stacking model that allows you to model weights conditional on pointwise covariates (Yao et al., 2022). Having stacking weights conditional on covariates is particularly useful in cases where we believe that models will perform better in different contexts. Common examples include the program's line of business or earned premium for the current accident year – different lines of business might be better explained by one model over another, and similarly for different accident years. Instead of simply choosing the best individual model for each context, stacking in such cases allows us to dynamically weight models when most appropriate.

For demonstration purposes, we will use the stacking model described by Yao et al. (2018), which estimates a single weight for each model. The model can be easily fit using `bayesblend` – an open-source `Python` library that we have developed that implements various different stacking models (Haines and Goold, 2024). To fit the stacking model, we first computed the posterior log likelihood values and posterior predictions for each datapoint in the test set as illustrated in Figure 5. We did so for both the random walk and mean reversion forecasting models. Note that each posterior log likelihood in the test set is derived based on the true loss, which the models are not trained on. We then used the posterior log likelihoods

and associated posterior predictions for each model as input to the `MleStacking` class in `bayesblend`, which outputs a blend of the posterior log likelihoods and predictions given the estimated model weights.

3.5.2 Model performance metrics

To assess model performance, we use a variety of out-of-sample metrics including (1) expected log pointwise predictive density (ELPD), (2) root mean squared error (RMSE), and (3) predictive versus true percentiles. The use of multiple metrics helps to hedge against the shortcomings of any particular metric, giving us a fuller understanding of how a model may perform on new programs. Note that all metrics are derived using the validation set as illustrated in Figure 5. Focusing on the validation set is important because we are most interested in future accident year performance. Additionally, it is necessary for a fair comparison between the individual models and the stacked model given that the latter is trained on the test set performance. Below, we define each metric before presenting the results.

As described in Section 2.3, ELPD is estimated using predictions on out-of-sample predictions. In our case, out-of-sample predictions for each triangle include those where $i = 2, ..., N$. However, because we are primarily interested in future accident year performance, we focus our metric definitions only on the "validation" datapoint, where $i = N = 10$ (see Figure 5). We define the validation set LPD for model as the log likelihood values for each validation datapoint, marginalized across the posterior samples S:

$$\begin{aligned} \text{LPD}_g^{\text{val}} &= \log p(\tilde{y}_{g10(10)} \mid \mathcal{Y}) \\ &= \log \int p(\tilde{y}_{g10(10)} \mid \theta) p(\theta \mid \mathcal{Y}) d\theta \quad \forall g = 1, ..., G \\ &\approx \frac{1}{S} \sum_{s=1}^{S} \log p(\tilde{y}^{(s)}_{g10(10)} \mid \mathcal{Y}) \end{aligned} \tag{11}$$

The expected LPD (ELPD) is then defined as the sum of LPD values across the validation datapoints, or across the different triangles g in our case:

$$\text{ELPD}^{\text{val}} = \sum_{g=1}^{G} \text{LPD}_g^{\text{val}} \tag{12}$$

Higher values of ELPD indicate better model performance. When comparing models, we took the difference in pointwise LPD values between models and then computed the sum of the differences and their associated standard errors as described by Vehtari et al. (2017). This pointwise comparison is akin to a paired t-test on the LPD values across triangles.

We defined pointwise out-of-sample RMSE as follows:

$$\text{RMSE}_g^{\text{val}} = \sqrt{\frac{1}{S}\sum_{s=1}^{S}(y_{g10(10)} - \tilde{y}_{g10(10)}^{(s)})^2} \quad \forall g = 1, ..., G \tag{13}$$

Unlike ELPD, RMSE primarily penalizes inaccuracy, where predictions further from the true loss are increasingly penalized. Similar to ELPD, for model comparison we computed the pairwise difference in RMSE across triangles, then computed the mean of these differences and their associated standard errors.

Finally, we computed the percentile of the true value in the posterior predictive distribution for each out-of-sample validation datapoint. Formally, the pointwise percentiles are defined as:

$$\text{PERC}_g^{\text{val}} = \frac{1}{S}\sum_{s=1}^{S}\mathbf{1}_{\{\tilde{y}_{g10(10)}^{(s)}, y_{g10(10)}\}} \tag{14}$$

where $\mathbf{1}$ is an indicator function that returns 1 if the true value is lower than the posterior sample and 0 otherwise:

$$\mathbf{1}_{\{X,Y\}} = \begin{cases} 1 & \text{if } X < Y \\ 0 & \text{else} \end{cases} \tag{15}$$

The resulting percentiles should follow a uniform distribution if the model's posterior predictions are well-calibrated. By assessing the empirical distribution of predictive percentiles, we can determine if the model is making loss predictions that are too low, too high, too certain, or too uncertain at the aggregate level.

3.5.3 Model performance results

Figure 6 shows how the random walk, mean reversion, and stacked models compare in terms of both ELPD and RMSE within the four lines of business in the Meyers dataset. There is significant variability in which model performs best across different lines of business, although the ELPD and RMSE metrics tend to agree with each other. Specifically, the mean reversion model tends to perform better for the private passenger auto (PP) and commercial auto (CA) programs, and the random walk model performs better for the general liability (OO) and worker's compensation (WC) programs. These findings align with our intuitive understanding of the different lines of business – auto programs tend to develop very quickly, meaning that the feedback loop between a management team observing losses and then

engaging in behaviors that change losses or premiums is rather short. Such dynamics are captured by the mean reversion mechanism in equation 7. Conversely, the losses underlying general liability and worker's compensation programs typically take many years to develop, making the aforementioned feedback loop longer and thus harder to act on. With enough historic data, it is possible to capture longer-term market cycle trends in such data, but the short term dynamics are better described by the random walk process in equation 6.

Across all lines of business, the stacked model is either similar to or better than the best individual candidate model within each line. The only exception is for general liability, where the ELPD metric slightly favors the random walk model over the stacked variant. Because the stacking model is trained on the test data and model performance is only estimated on the validation data (see Figure 5), it is not surprising that the stacked model occasionally performs worse than individual candidate models. However, when looking across lines, the stacked model provides the best compromise in the sense that it is the model configuration that performs best in aggregate across contexts.

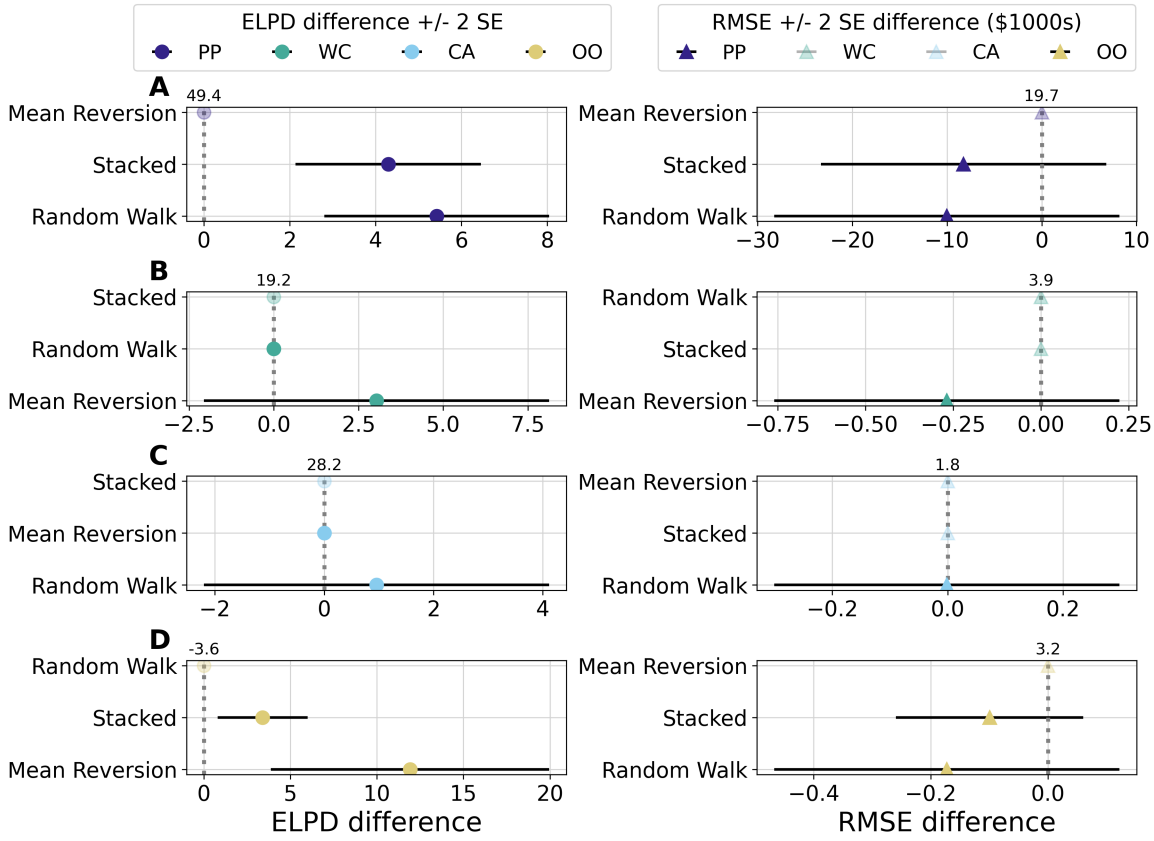


Figure 6: ELPD (left) and RMSE (right) differences (+/- 2 standard errors on the difference), ordered by Validation dataset performance for each model and line of business. The best-performing model is the reference model, shown at the top of each panel. Absolute ELPD and RMSE values are displayed above. Positive ELPD differences with an uncertainty interval that does not cross zero indicates a credible difference at the 95% level in favour of the top model. Negative RMSE differences with an uncertainty interval that does not cross zero indicates a credible difference in favor of the top model. Note that in cases where non-reference models do not have visible intervals, the uncertainty intervals are just too narrow to be visible on the plot.

Figure 7 shows the model calibration results for each model within each line of business. Here, we see that calibration is good for commercial auto and worker's compensation lines, but calibration for private passenger auto and general liability lines reveals that loss predictions have an upward bias relative to true losses across all models. We tried various different modifications to the underlying models in an effort to improve calibration for these lines, including changes to the loss development training windows τ and ρ, different measurement error assumptions in the forecasting model (e.g., normal as opposed to lognormal error), changes to priors for both loss development and forecasting models, and different outcome distributions for the loss development and forecasting models (e.g., gamma as opposed to lognormal) – all changes resulted in the same general pattern found in Figure 7.

Poor calibration should be taken quite seriously, as having accurate forecasts with proper uncertainty quantification is crucial for pricing in ILS deals. Therefore, the poor calibration for private passenger auto and general liability lines would indicate that the models presented are not suitable for production use in those lines, even if they perform well according to other tests. Of course, these particular results are highly dependent on the data used for backtesting purposes. The Meyers dataset is limited in that we only have a 10-year window of history for each underlying triangle, starting in 1988 and ending in 1997. This limited history means that all out-of-sample validation data is coming from accident year 1997 (see Figure 5), which may not give a generalizable view of how models should be expected to perform for arbitrary accident years. In fact, exogenous factors such as the macroeconomic environment or general industry conditions – neither of which are explicitly captured by models presented here – could induce correlations in ultimate losses across programs within each line of business. When left unaccounted for, such correlations could contribute to poor calibration.

In practice, uncertainty regarding the generalizability of any given accident year is why we typically perform backtests using a rolling window approach, where the procedure in Figure 5 is iterated across overlapping 10-year accident year increments. However, this rolling window approach requires a dataset with much more data, and such rich datasets are not openly available in the insurance industry. Anecdotally, using our internal licensed data with much more history, we have found that calibration for the forecasting models presented here is generally good across lines of business, suggesting that the results in Figure 7 may in part be due to the idiosyncracies of accident year 1997.

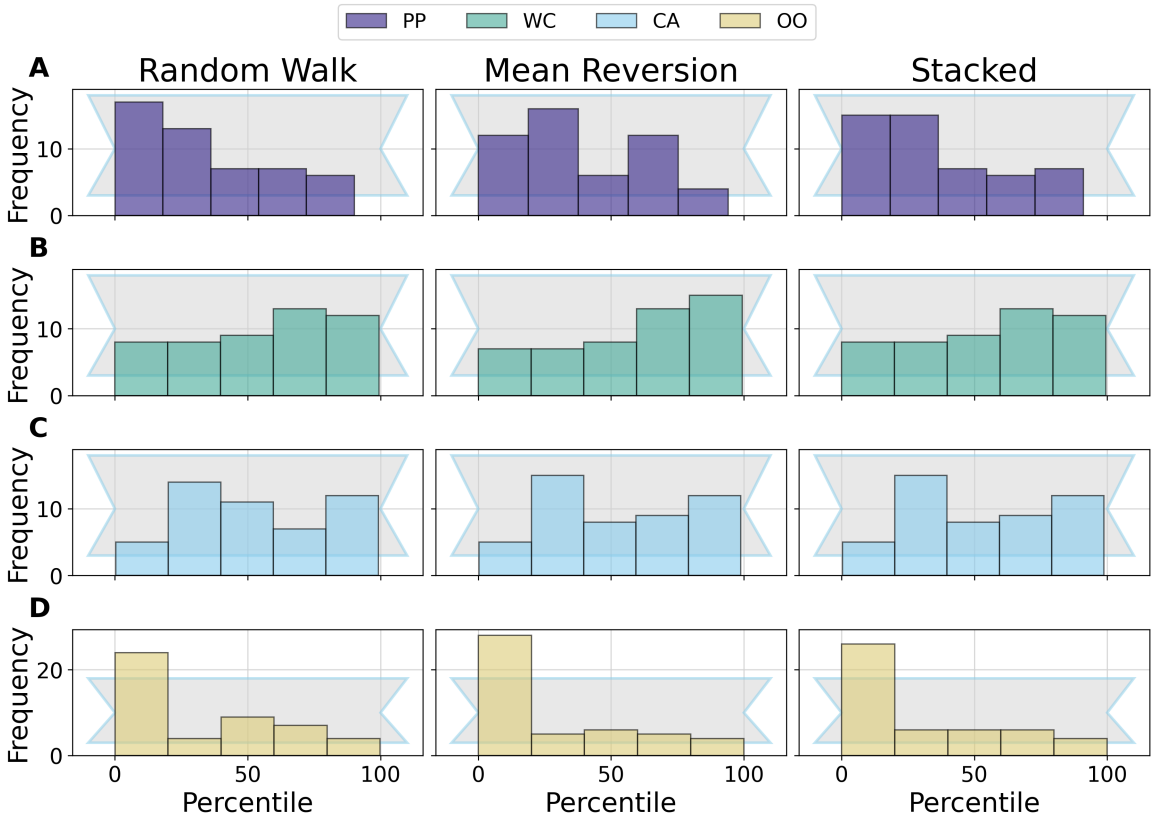


Figure 7: Percentiles of the true hold-out Validation dataset losses within the corresponding posterior predictive distributions for each model and line of business (panels A through D). Grey shaded regions provide the 99% intervals of a target uniform distribution, for reference. Right-skewed histograms indicate under-estimation, left-skewed histograms indicate over-estimation, inverted-U histograms indicate predictions that are uncertain, and U histograms indicate predictions that are too certain.

3.6 From Forecasts to Cashflows

Once we have selected a model configuration that can forecast ultimate losses for future accident years to a desired level of accuracy (including calibration, RMSE, etc.), the next step is to estimate cashflows that can be used to understand the profitability of the insurance program. There are two types of cashflows that are relevant for this step: (1) the "underwriting" cashflow, which captures primarily how losses are paid out across time for the future accident year being securitized; and (2) the "investment" cashflow, which captures interest on invested premium dollars (i.e. the "float") as well as bespoke deal structure terms (collateralization schemes, profit commission to cedants, etc.). For the current demonstration, we will focus primarily on the underwriting cashflow, as the investment cashflow is a deterministic function of underwriting cashflow (i.e. it is mostly an exercise in accounting) so is outside the scope the current work.

3.6.1 Estimating future underwriting cashflows

There are many ways to estimate future underwriting cashflows, but for illustration we will take a simple approach here. Underwriting cashflows are comprised primarily of paid losses (outflow) versus premiums collected (inflow). Other inflows can include reinsurance or legal recoveries, and other outflows can include expenses related to claim settlement or management – here we will only consider paid losses and premiums.

Given the estimated age-to-age factors and the next year(s) ultimate loss estimate(s) from the development and forecasting models, respectively, we can walk back from the ultimate loss estimate to obtain estimates of future paid losses. Because we have full posterior distributions for the ultimates loss estimates and age-to-age factors, we can propagate both parameter and epistemic uncertainty from the development and forecasting models into the paid loss estimates by leveraging the MCMC samples. Assuming $\tilde{r}_{i\infty} = \tilde{r}_{i(10)}$ represents the ultimate loss ratio prediction for future accident year i obtained from a forecasting or stacking model, which makes $\tilde{y}_{i\infty} = \tilde{y}_{i(10)}$ the corresponding ultimate loss estimate derived by multiplying the loss ratio by the premium associated with the same period, we can estimate future paid losses as follows:

$$
\begin{aligned}
\tilde{y}^{(s)}_{i(9)} &= \tilde{y}^{(s)}_{i(10)} / \alpha^{(s)}_{9} \\
\tilde{y}^{(s)}_{i(8)} &= \tilde{y}^{(s)}_{i(9)} / \alpha^{(s)}_{8} \quad \forall s = 1, ..., S \\
&\vdots \\
\tilde{y}^{(s)}_{i(1)} &= \tilde{y}^{(s)}_{i(2)} / \alpha^{(s)}_{1},
\end{aligned}
\tag{16}
$$

where $\alpha^{(s)}_{j}$ is the sth posterior sample of the jth age-to-age factor. The result is a full posterior distribution of paid loss estimates for each future accident year i (see Figure 8). Across development, these paid loss estimates, relative to premiums, correspond to the underwriting cashflow associated with the future accident year.

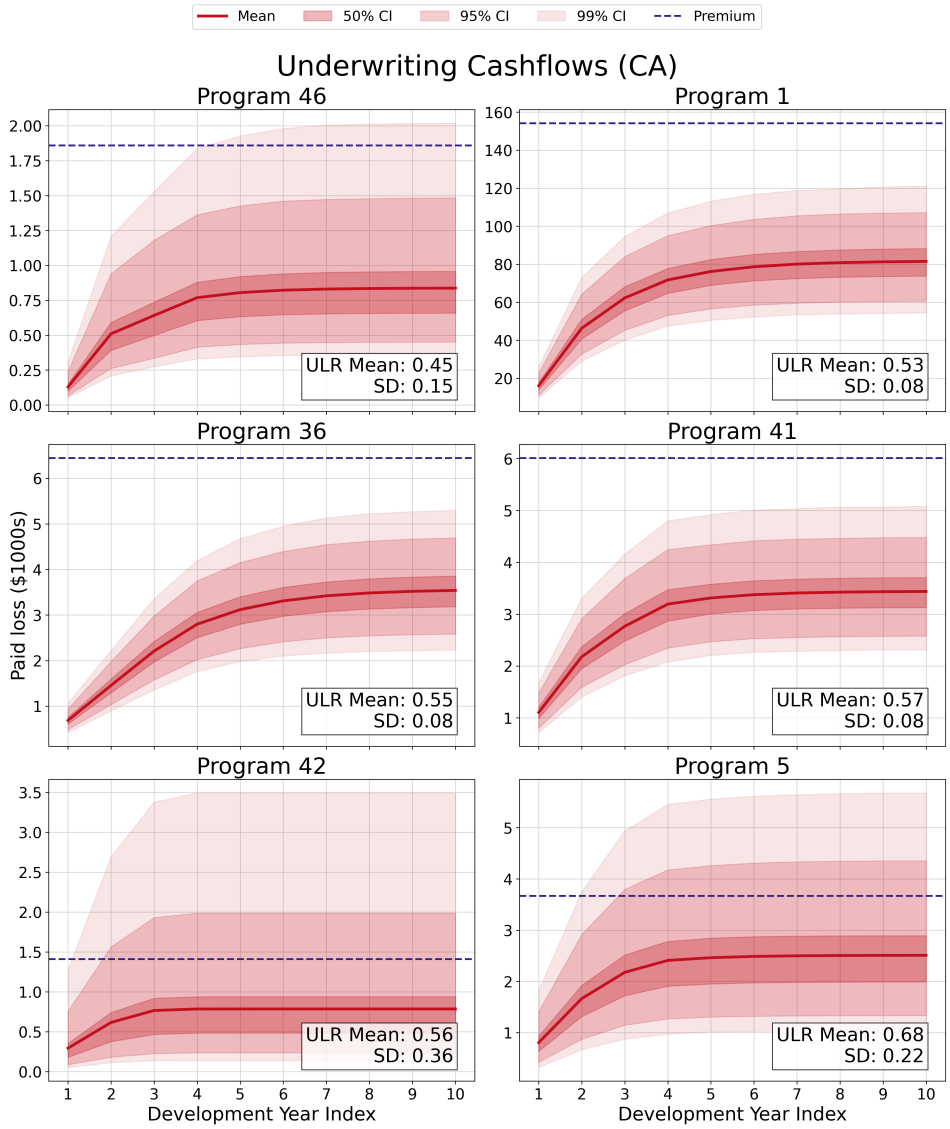


Figure 8: Underwriting cashflows for future accident years based on historic development patters and forecasted ultimate losses for a random selection of commercial auto programs in the Meyers data. Results are based on the development and forecasting models fitted during the backtests described above. Underwriting cashflows were estimated per the walk-back method described in equation 16 using random programs selected from the Backtest workflow (see Section 3.5) illustration. Forecasted ultimate losses were based on the stacked predictions from the random walk (equation 6) and mean reversion (equation 7) models, and age-to-age factors were obtained from the chain-ladder (equation 4) and generalized Bondy (equation 5) models fitted to each program.

3.6.2 Underwriting to investment cashflows

Because underwriting cashflows only inform us as to how insurance policies are expected to be paid out over time, they are not sufficient for pricing ILS deals. An ILS fund can be thought of as a trust account comprised of several additional inflows and outflows that determine its overall performance. In addition to the inflows and outflows associated with underwriting cashflows, the primary investment cashflows include investment income and initial investor capital (inflows) as well as expenses associated with managing or brokering the security (outflows). Although we will not cover technical details on investment cashflows here, it is worth discussing the general structure of an ILS fund to understand the relationship between underwriting and investment cashflows.

Initially, the fund consists of the total premiums collected from policyholders plus the investor capital. The capital requirement is determined as some proportion of the expected claim payments, ensuring that the fund has sufficient reserves to cover potential losses. Notably, the investor capital is considered leveraged because interest is earned on the total amount of investor capital plus the premiums collected. Because the capital contributed by the investor is often significantly less than the amount of premium ceded by the insurer, it is common for investors to receive more than 2x leverage on investment returns. This leverage means that even if the interest is a low single-digit yield, such as from bonds or treasury bills in the current market, investors often realize a much higher internal rate of return (IRR) on capital. The longer it takes for claims to come in and be paid out (i.e. the longer it takes for losses to develop), the longer the investor's capital remains leveraged.

Of course, many of the above details are negotiable and depend on the specific deal structure. For example, the capital requirement may be higher or lower if the underlying insurance porfolio is concentrated or diversified, respectively. Similarly, expenses can vary widely depending on the complexity of the deal and the services provided by the ILS fund manager. For example, managers of the insurance program may be incentivised with bonuses based on the loss ratio of the program, affecting underwriting cashflows. Additionally, the ILS broker may receive a fee based on the total amount of capital raised, affecting investment cashflows.

4 Working example

In this section, we apply the full Bayesian workflow presented in Section 3 to the Swiss private liability loss development triangle from Balona and Richman (2020) (refer to Tables 23-24). These data are originally from Gisler (2009), but Balona (2020) add simulated premium data to the triangle that is needed for our workflow, so we will refer to Balona and Richman (2020) as the source from hereon out. The Swiss private liability triangle has 18 years of history spanning from 1979 to 2015, which originate from a major Swiss insurer. Note that the exact figures have been adjusted for privacy purposes–we report values directly

from Balona and Richman (2020) in thousands of dollars, but emphasize that the scaling is arbitrary. As seen in Figure 9, the data exhibit the typical long-tailed, early-lag volatility that characterizes many insurance programs undergoing casualty securitization. As such, it makes for an insightful example of how to apply the Bayesian workflow in practice.

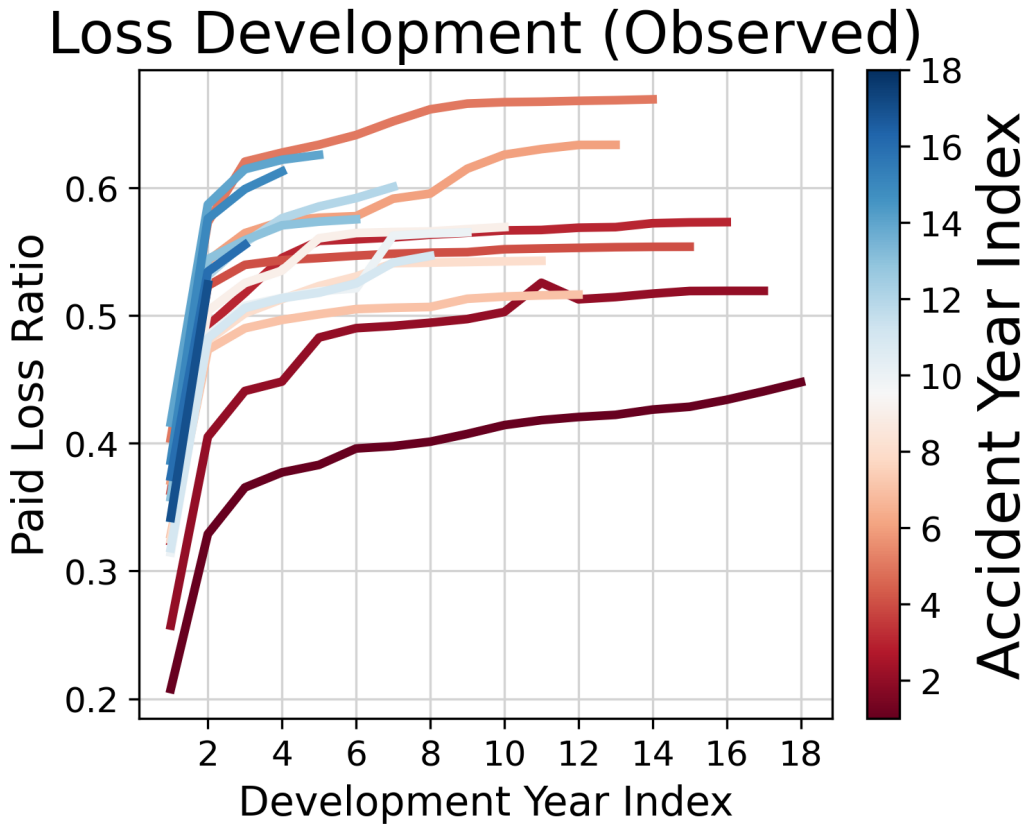


Figure 9: Observed loss ratio development patterns from the Swiss liability program in Balona and Richman (2020), indicating a generally long-tailed structure with variability in early development years.

4.1 Loss development

Following the historical loss development workflow from Section 3.1, we separate loss development into a "body" process (via the Bayesian chain-ladder model in equation 4) and a "tail" process (via the generalized Bondy model in equation 5). Because private liability falls under the more general umbrella of general liability insurance, we specify the the chain-ladder and generalized bondy training thresholds using similar values as for those used to backtest the general liability line above (see Section 3.5). Specifically, we set $\tau = 6$ and $\rho_1, \rho_2 = (4, 15)$. We also used the same prior distributions as specified in equations 4 and 5 for the chain-ladder and generalized Bondy models, respectively.

Note that the upper threshold we set for the tail model (ρ_2) is higher here because the Swiss liability triangle has 18 years of history available, whereas the Meyers data used for backtests only had 10 years of history. Further, unlike for the backtest, we do not set the upper tail threshold to the maximum possible given the training data (i.e. $\rho_2 = 18$). Including the full training data means that tail development factors for the most recent development lags are informed only by the most historic accident years where data are available. By not fitting the tail model to the most recent development lags, we allow for the model to be informed only by development patterns where we have multiple accident years, which can make the tail estimates more robust to noise that may otherwise be present in 1 or 2 early accident years.

Figure 10 shows posterior predictions from the development models on the Swiss private liability program, faceted by accident year to better visualize the uncertainty in the predictions. Generally, we see that the model predictions are mostly flat after 6-7 years of development. It is also clear that the model is more uncertain for more recent accident years, which is expected given we only have one or more observed years of development for the latest accident years. Overall, the importance of the development model is most readily observed for later accident years, where the model captures the expected increase in cumulative losses across early development periods before converging toward the ultimate loss ratios for each year.

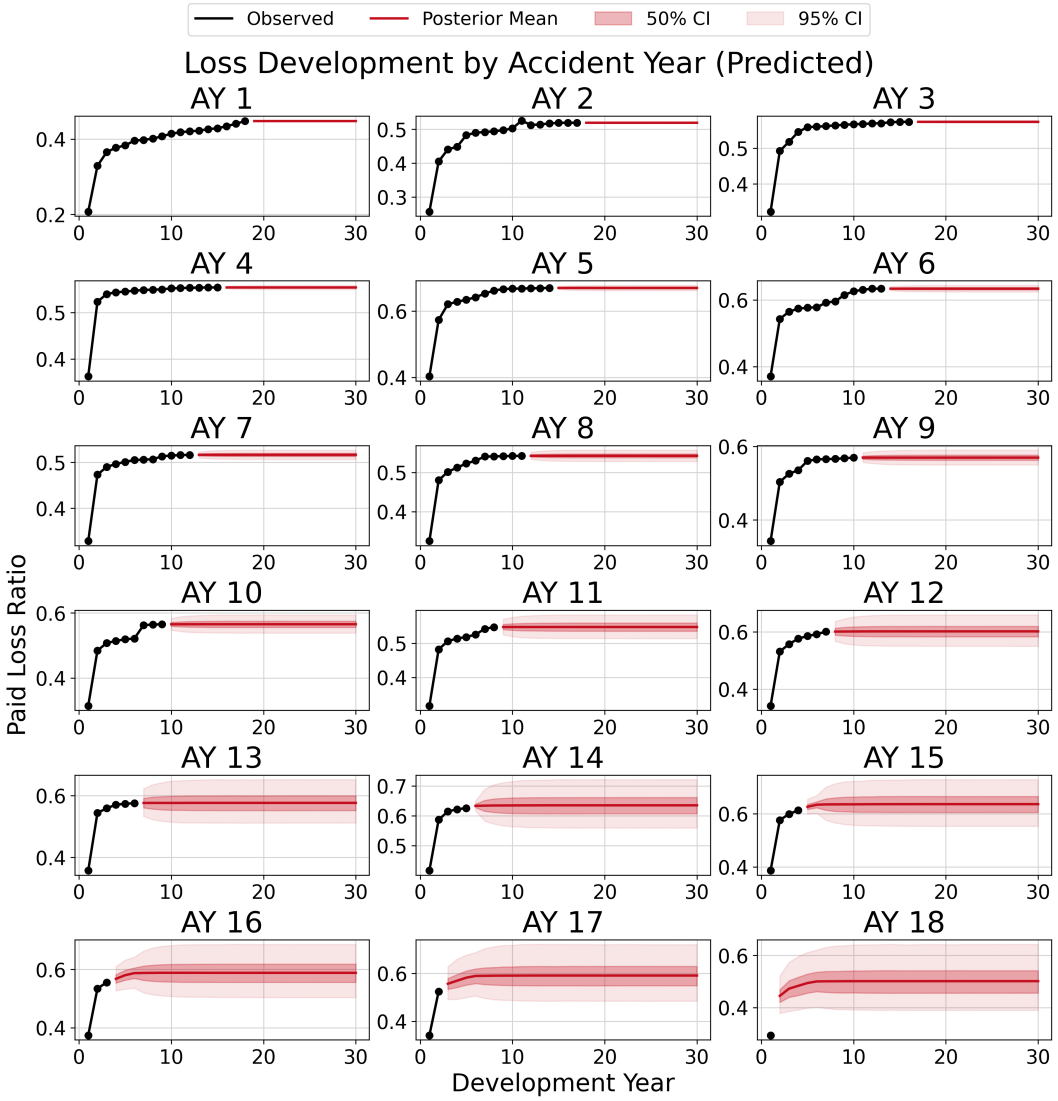


Figure 10: Observed loss ratio development patterns along with posterior predictions from loss development models for the Swiss liability program in Balona and Richman (2020). The predicted ultimate loss ratio means and standard deviations (i.e. predicted means and standard deviations at development year 30) are used as input to the forecasting model described in Section 4.2. The shaded regions correspond to the 50% and 95% credible intervals for the predictions.

4.2 Ultimate loss forecasting

Given the predicted ultimate losses obtained for historic accident years (see Figure 10), the next step is to predict ultimates for future accident years. For the current example, we will use the same forecasting models as described in Section 3.2. Specifically, we fit both the random walk (equation 6) and mean reversion (equation 7) models, both with the

measurement error specification in equation 8. After generating one-year-ahead predictions from each fitted model, we then blend together the posterior predictive distributions to arrive at the final one-year-ahead ultimate loss ratio prediction.

For blending the forecasting model predictions, we used the same `MleStacking` model as in our backtests (see Section 3.5.1). Recall that in our backtests, we trained a separate stacking model for each of the four lines of business in the Meyers (2015) dataset. To blend the Swiss liability triangle predictions, we used the stacking model trained on the general liability subset of the Meyers data. The `MleStacking` model only assigns a single static weight to each forecasting model, which in our case produced weights of 0.55 for the random walk model and 0.45 for the mean reversion model. Given these weights, the blended ultimate loss ratio distribution is generated by sampling the posterior predictive distributions from each forecasting model in proportion to each model's respective weight.

Finally, for our forecasting model priors, we used data-driven priors as described in Section 3.3. Specifically, when conducting the backtests for each line of business in the Meyers dataset (see Section 3.5), we saved the posterior distributions from the hierarchically estimated forecasting models (see equation 10), summarized the group-level parameters for each line of business, and used these summaries as the priors for each forecasting model fitted to the Swiss liability triangle:

$$
\begin{aligned}
\eta_{\text{init}} &\sim \mathcal{N}(-0.92, 0.41) \qquad \text{(Random Walk)} \\
\epsilon &\sim \mathcal{N}(-2.17, 0.22) \\
\gamma_1 &\sim \mathcal{N}(-2.18, 0.96) \\
\gamma_2 &\sim \mathcal{N}(-2.13, 0.92)
\end{aligned}
$$

$$
\begin{aligned}
\eta_{\text{init}} &\sim \mathcal{N}(-1.49, 0.86) \qquad \text{(Mean Reversion)} \\
\epsilon &\sim \mathcal{N}(-2.66, 0.15) \\
\mu &\sim \mathcal{N}(-0.65, 0.52) \\
\phi &\sim \mathcal{N}(1.27, 0.95) \\
\gamma_1 &\sim \mathcal{N}(-2.23, 0.91) \\
\gamma_2 &\sim \mathcal{N}(-2.24, 0.86)
\end{aligned}
$$

Figure 11 shows the posterior predictive distributions for each forecasting model across accident years, including the one-year-ahead predictions. The models capture the same general patterns, although the mean reversion model tends to produce predictions with less uncertainty than the random walk model. This occurs because the mean reversion process pulls year-to-year predictions toward the mean of the historical data, which reduces overall variability in the predictions. Further, the mean reversion model produces a slightly higher ultimate loss ratio mean than the random walk model. The difference in both the expectation and uncertainty of predictions across models highlights the importance of model blending, allowing us to combine models as opposed to selecting just one.

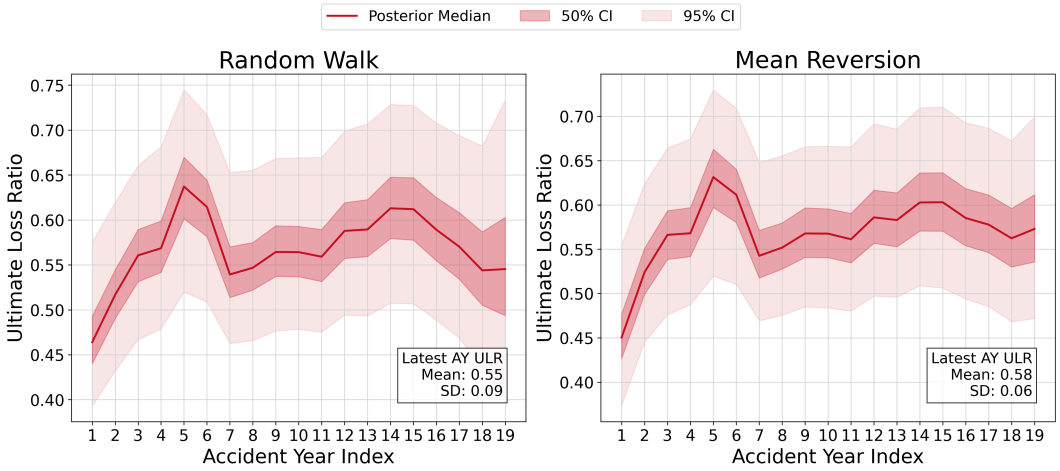


Figure 11: Posterior predictive distributions for each forecasting model across accident years for the Swiss liability triangle from Balona and Richman (2020). The shaded regions correspond to the 50% and 95% credible intervals for the predictions.

4.3 Underwriting Cashflows

The final step we will cover in our working example is to predict one-year-ahead underwriting cashflows given the forecasted ultimate loss ratio predictions and the historic loss development patterns. Here, we use the walk-back method outlined in equation 16. As explained in Section 3.6.1, the walk-back method starts from the one-year-ahead ultimate loss ratio obtained from the forecasting models, then uses the age-to-age factors from the loss development model (Section 4.1 above) to iteratively predict back to the first development lag. This process results in a full time-series of posterior predicted cashflows for the one-year-ahead accident year, which represent the paid losses expected to emerge for the future accident year across time.

Figure 12 shows the predicted underwriting cashflows for the Swiss liability triangle. It is clear from Figure 12 that paid losses are expected to converge to the ultimate loss after approximately 6 years, with an expected loss ratio of 56% and a 97.5%ile loss ratio of about 72%. Given the relatively low expected ultimate loss ratio along with the development pattern, this program is a good candidate for securitization (see Section 3.6.2).

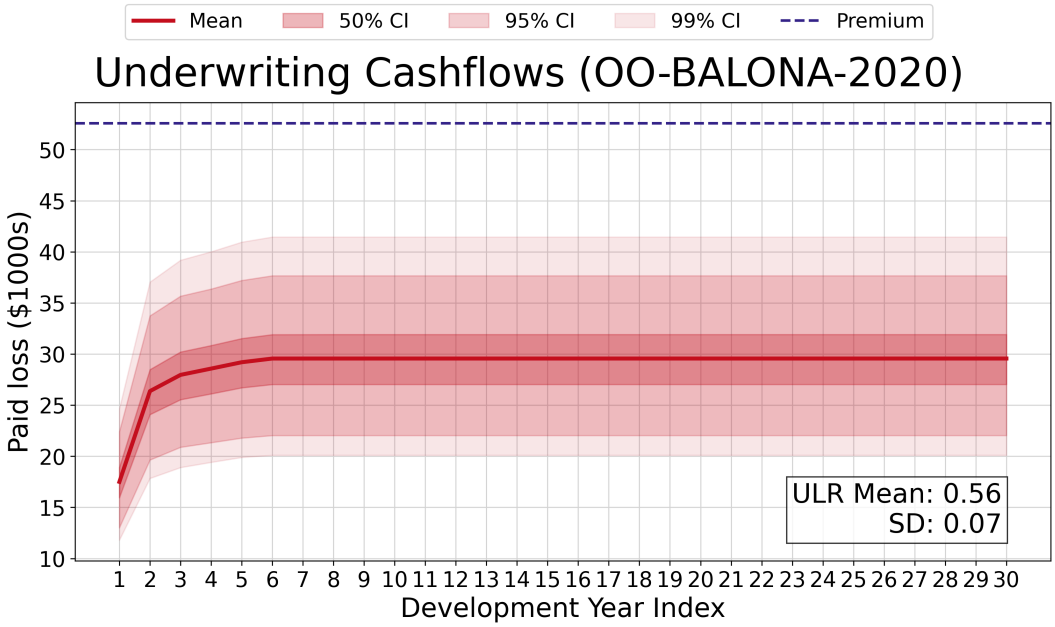


Figure 12: Predicted paid loss development for the one-year-ahead accident year for the Swiss liability triangle from Balona and Richman (2020). Paid losses for future accident years represent the expected underwriting cashflows associated with the ILS deal. The shaded regions correspond to the 50% and 95% credible intervals for the predictions.

5 Discussion

The Bayesian workflow presented in this paper offers a comprehensive framework for approaching the first hurdle necessary to securitize casualty insurance risk – obtaining accurate, well-calibrated ultimate loss predictions for future accident years. Our simulations and real-world applications demonstrate that the Bayesian framework is not only powerful enough to account for complexities underlying casualty ILS, but that it is also maximally transparent. Specifically, the use of Bayesian models requires us to encode each of our assumptions about loss development and forecasting dynamics into a mathematical expression that can be thoroughly inspected and critiqued. Cross-validation (i.e. backtesting) then shows us whether our model is expected to perform well enough for real-world applications. We believe these methods are crucial for ensuring that casualty ILS deals are priced in a data-driven way, which increases transparency and trust between insurers, captial providers, and analysts alike.

There is still much work to be done in the casualty ILS space both on loss development and loss forecasting. For example, for loss development, we have found that the selection of which development lags should be used for training the body versus tail models (i.e., τ and ρ in equations 4 and 5, respectively) can have a large influence on ultimate loss predictions. This finding inspired work by Goold (2024), who showed that hidden markov models can

be used to model how the loss development process switches from the body to tail state as development progresses. Despite showing similar or better performance on out-of-sample tests when compared to using traditional training window methods, both the hidden markov and traditional approach resulted in relatively poor calibration, emphasizing the need for further research on the core assumptions underlying common loss development models.

Similarly, for forecasting models, how uncertainty is propegated from the loss development stage into the forecasting stage has strong implications for the forecasted ultimate losses. In the workflow we presented, we took an error-in-variables approach (see equestion 8). However, there are multiple other reasonable approaches to take, the most obvious being jointly modeling the loss development and forecasting stages. Historically, we have avoided doing so for practical reasons as mentioned in section 3.2.3. However, we believe this could be a good area for future research – the inconvenience of the joint model could be outweighed by potential performance improvements. Further, ultimate loss ratios from year-to-year can be highly impacted by general market conditions (Berger, 1988), leading to market cycles that have an observable impact on loss ratios. Models that can leverage market cycle information therefore have strong potential to improve ultimate loss forecasts. Our Bayesian framework is well suited to test if such models actually lead to meaningful improvements in accuracy (or calibration) over models that do not account for industry dynamics.

More generally, this is the first paper of its kind to introduce a fully worked-out Bayesian workflow to address casualty insurance loss modeling. In the same way that models used throughout actuarial science have been studied, extended, and critiqued, we believe that actuarial workflows should be formalized and rigorously studied in an effort to both improve and disseminate them. To support future research in this vein, we have made all of the scripts used throughout this paper available in a repository (`https://github.com/LedgerInvesting/bayesian-workflow-paper-2024`). In doing this work, it has also become clear to us that future research on ILS (and throughout actuarial science more generally) would greatly benefit from the availability of richer open datasets that can be used to test model assumptions. Standardized open datasets would allow for researchers across institutions to compare models against common benchmarks, accelerating model developent throughout actuarial science. The Meyers (2015) dataset is a great starting place, but any results based on it should be taken with a grain of salt given its size and lack of recency. Similarly, the Swiss liability triangle from Balona and Richman (2020) is a good candidate for testing model performance given its 18 year history. However, it is just a single triangle, and the most recent accident years available are from 2015. Larger datasets with longer, more recent histories and more lines of business are needed to ensure that published backtest results are generalizable. Although many private organizations have access to such data (e.g., through purchase agreements with data aggregators), licensing restrictions prohibit open use of these datasets. The impetus then is on the research community to advocate for more data sharing and collaboration, which could ultimately lead to more robust and reliable models in the field.

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